

Why Sorting Students Is Not Enough

Evidence from Three Ability-Grouping Experiments

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Full Draft Available on Request

1 Introduction

Many organizations must assign people to tasks before they know the match perfectly. A firm uses credentials to assign workers to jobs; a hospital uses triage information to assign patients to units; a lender uses credit scores to assign borrowers to contracts. Schools face the same problem. They must decide what instruction a child should receive, and they usually make that decision using grade.

Grade is an administratively simple assignment rule. It groups students by age and curricular exposure and coordinates classroom instruction with the formal curriculum. But grade is a proxy for readiness. Human capital theory treats learning as the product of prior skills and instructional inputs, not age labels (Ben-Porath, 1967); dynamic models of skill formation sharpen the point because early mismatch can lower the return to later instruction (Cunha and Heckman, 2007). Grade is informative when progression through school is tightly linked to learning. It becomes less informative when late enrollment, repetition, automatic promotion, and unequal home inputs weaken that link.

That link is often weak in low-income school systems. Within-grade achievement variance is large, adjacent grades overlap substantially in skill, and many students reach upper primary without mastering foundational literacy or numeracy (Banerjee et al., 2007; Muralidharan et al., 2019). A single Grade 3 classroom may contain students who cannot read a sentence and others who can read fluently; in mathematics, some students may still be learning to count while others are ready for multiplication. In such classrooms, grade is already a form of sorting. The question is whether another signal changes instructional match relative to grade.

One response is cross-grade ability grouping. The Joplin Plan, introduced in the 1950s, re-

grouped elementary students across grade lines for reading, so that instruction could be pitched to measured reading level rather than age-grade placement (Floyd, 1954). The idea is close to the instructional-level tradition in educational psychology: students learn most from content that is neither trivial nor out of reach (Betts, 1946; Vygotsky, 1978). Cross-grade grouping expands the assignment margin relative to within-grade tracking without requiring more teachers or classrooms. It changes the allocation of students to instruction.

The historical evidence provides a benchmark. In a best-evidence synthesis, Slavin (1987) reports a median effect size of $+0.45$ SD for the Joplin Plan, and a related review of nongraded elementary schools finds similarly large gains for one-subject cross-grade grouping (Gutiérrez and Slavin, 1992). These are older studies, so they should not be read as modern experimental estimates. Those studies also describe the operating conditions under which grouping was implemented. Slavin (1987) finds that grouping works best when teachers can respond to the new classrooms: grouping is limited to one or two subjects, assignments are reassessed frequently, and teachers vary level and pace in response to student needs. In those studies, the Joplin Plan combined sorting with adaptation.

This paper studies a version of cross-grade grouping in which that adaptation channel is more limited. The experiments take place in scripted-instruction programs, where lesson content is fixed by track-specific scripts. In the standard tracking model, as in Duflo et al. (2011) and the classroom-production logic of Lazear (2001), tracking can raise achievement because a more homogeneous classroom lets the teacher adjust content, pace, effort, or classroom management. Scripted instruction changes the technology. If the teacher follows a fixed sequence of lessons, grouping works only if the assignment rule sends students to content consistent with their preparedness and the school actually delivers that content.

The empirical question is therefore not whether ability grouping has positive effects in every form. The Joplin evidence and the modern Country A tracking experiment show positive effects in settings with teacher discretion. The question is whether the same organizational form works when the production technology shifts from teacher adaptation to assignment to scripted content. This shift changes which inputs determine the realized effect. A homogeneous classroom can still be taught at a level that does not match student preparedness. A placement assessment can predict readiness less accurately than grade. A school can form the groups on paper but fail to maintain

them in classrooms. And even when groups are formed correctly, the assigned curriculum may not be taught.

We separate these requirements into three inputs. The first is the quality of the placement signal: whether the assessment predicts the skills that matter for later learning. This is related to the psychometric idea that a test’s value depends on its reliability and validity for the intended use (Lord and Novick, 1968; Cronbach and Meehl, 1955). The second is assignment execution: whether intended groups are formed and maintained. The third is delivery of the assigned content. Recent work on external validity emphasizes that a null realized effect need not imply a null latent effect when implementation is incomplete (Angrist and Meager, 2023). In this setting, implementation is not a single dose. A more predictive assessment has little value if students are not assigned according to it, and assignment according to the assessment has little value if the curriculum tied to that assignment is not delivered.

We study three randomized evaluations of cross-grade ability grouping in scripted-instruction programs. The experiments span Country A, Country B, and Country C; literacy and numeracy; and two- and three-track designs. The broad instructional architecture is similar across settings, but the country cases differ sharply in placement-signal quality, group execution, and delivery of assigned content. For that reason, the paper does not treat the pooled effect as the main object. The interpretation comes from the country-specific pattern.

The experiments do not produce a positive average effect, but the failures are different. Country B is the low-signal case. Assignment was mechanically executed, but the baseline assessment explains less than 7% of endline variation ($\rho = 0.055$). The intervention therefore replaced grade with a placement rule with low predictive content. The estimated effect is -0.21 SD ($p = 0.12$), with harm concentrated in the upper track. The Country B result comes from executing that rule, not from partial implementation of a rule whose predicted match exceeds grade.

Country A isolates the sorting channel more directly. The placement assessment is highly predictive ($\rho = 0.53$), assignment execution is complete ($\omega = 1$), and the intervention sharply compresses within-class dispersion. It also lowers predicted instructional mismatch relative to grade assignment. Yet the average effect is near zero (0.031 SD, $p = 0.56$). Students with larger predicted assignment gains do not benefit more, and movers do not outperform stayers. Country A is inconsistent with a story in which improved sorting alone is sufficient. In this scripted setting,

sorting at observed delivery does not translate into learning.

Country C was designed as the high-input case. Country C was chosen because general delivery quality was high, and the experiment was designed as a three-track numeracy intervention. In practice, the top group was thin, so the preferred reduced-form specification collapses Country C to the support-aware contrast between lower and higher groups, with unrestricted three-group estimates in the appendix. The higher group has a positive T3 endterm effect of +0.236 SD (SE = 0.132) and a stacked endterm effect of +0.214 SD (SE = 0.095). But implementation records show that the high-input design was not implemented as specified. Assignment records are missing for about one-third of treated students, many schools failed to maintain the group structure, teachers often delivered content for the wrong track, and the assignment-linked numeracy curriculum was delivered in only 26% of scheduled lessons. The estimated assignment-execution primitive is $\omega = 0.84$ and delivery fidelity is $\tau = 0.31$.

The structural exercise summarizes the missing case that none of the experiments delivers directly: a placement rule that improves on grade, near-complete group formation, and high delivery of assigned content. The preferred benchmark predicts an ATE of about +0.19 SD. The benchmark is used for scale; the sensitivity exercises point to the threshold logic. Raising assignment execution alone has limited predicted payoff when delivery is low or when the placement rule does not improve match. Raising predictive content alone has limited predicted payoff when the assigned curriculum is not delivered. In this technology, information, assignment, and delivery are complements.

The paper’s closest antecedents are studies of tracking and ability grouping. Meta-analyses of within-grade tracking generally find small or mixed average effects (Slavin, 1987, 1990; Kulik and Kulik, 1987; Terrin and Triventi, 2022), with important heterogeneity across tracks and institutional settings (Burke and Sass, 2013; Hoxby and Weingarth, 2005; Antecol et al., 2016; Garlick, 2018; de Roux and Riehl, 2022). In developing countries, the main modern benchmark is Duflo et al. (2011), who find gains from within-grade tracking in Country A under teacher discretion. Recent evidence from other settings similarly suggests that grouping can raise achievement for some students while leaving open why effects differ across environments (Busso and Frisanchi, 2023). This paper studies a different margin: cross-grade grouping, the form historically associated with the Joplin Plan (Floyd, 1954; Slavin, 1987; Gutiérrez and Slavin, 1992).¹ This margin differs from

¹There is also a large body of research on between-school ability grouping, mainly from Europe where tracking

a more intensive version of within-grade tracking. It replaces grade with a placement signal, so the economic question is whether the new signal predicts instructional match relative to the rule already in use.

The equity literature raises a different concern. Work in sociology and education emphasizes that tracking can stigmatize lower-track students, reduce teacher expectations, and concentrate weaker instruction in lower groups (Oakes, 1985; Gamoran, 1992; Gamoran and Nystrand, 1994). These concerns describe channels through which the production function can change when students are grouped. In the setting studied here, negative effects arise when the placement signal has low predictive content, when assignments are not maintained, or when lower groups receive mismatched content. The same assignment policy can therefore harm students in one institution and raise learning in another without implying that instructional match has the same value everywhere.

The mechanism also connects to work on educational production, peer assignment, targeted instruction, and implementation. In Lazear (2001), students share teacher time, pace, and classroom order, so peer composition and instructional response enter the classroom production function. Tracking experiments usually move peers, ranks, teacher behavior, and content together (Hoxby and Weingarth, 2005; Carrell et al., 2013; Vigdor and Nechyba, 2007; Rjosk, 2022). Scripted instruction narrows the empirical object: can the school use a signal to match students to lessons they are ready to learn? Targeted-instruction interventions show the value of aligning content to assessed learning levels (Banerjee et al., 2017; Duflo et al., 2020; Muralidharan et al., 2019). Success for All combines structured reading instruction, frequent assessment, and cross-grade regrouping in a form related to the Joplin Plan (Cheung et al., 2021), and its implementation literature shows that teacher beliefs, adaptations, and local take-up shape delivery (Datnow and Castellano, 2000). In our setting, the placement test and the content assignment are explicitly linked. A placement test is not targeted instruction. It becomes targeted instruction only if the test predicts the decision-relevant skill, the organization acts on the signal, and the assigned content is taught.

The measurement and implementation literatures clarify why the signal cannot be separated from its use. A test score is useful only for a particular decision. Reliability and validity therefore matter not as abstract properties of a score, but because they determine whether the score can

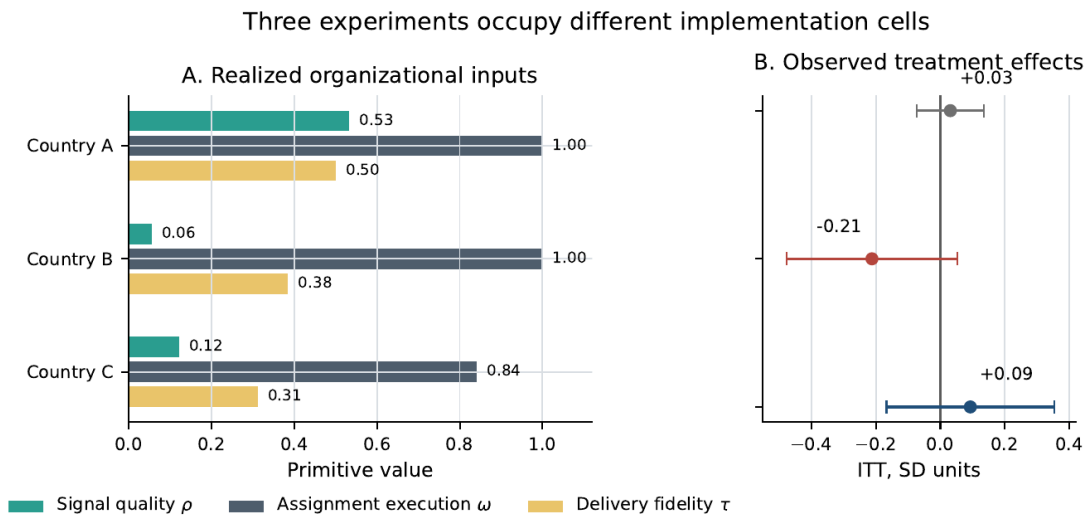
between academic and vocational tracks is prevalent (Fu and Mehta, 2018; Lavrijsen and Nicaise, 2015). We focus on within-school ability grouping.

support the assignment the school wants to make (Lord and Novick, 1968; Cronbach and Meehl, 1955). Angrist and Meager (2023) show that null realized effects need not imply null latent effects when implementation is incomplete or unobserved. We build on that point by showing why allocation policies require more than a scalar implementation adjustment. Faithfully executing a signal that worsens match can reduce learning, and an informative signal has little value unless the organization can act on it and deliver the corresponding input.

Section 2 formalizes the assignment problem. Section 3 describes the three experiments and the institutional differences that shape implementation. Sections 4–5 report the reduced-form effects and reallocation diagnostics. Section 6 estimates the benchmark for the missing high-input case. Section 7 returns to the tracking debate and the policy question of when an assessment should replace grade.

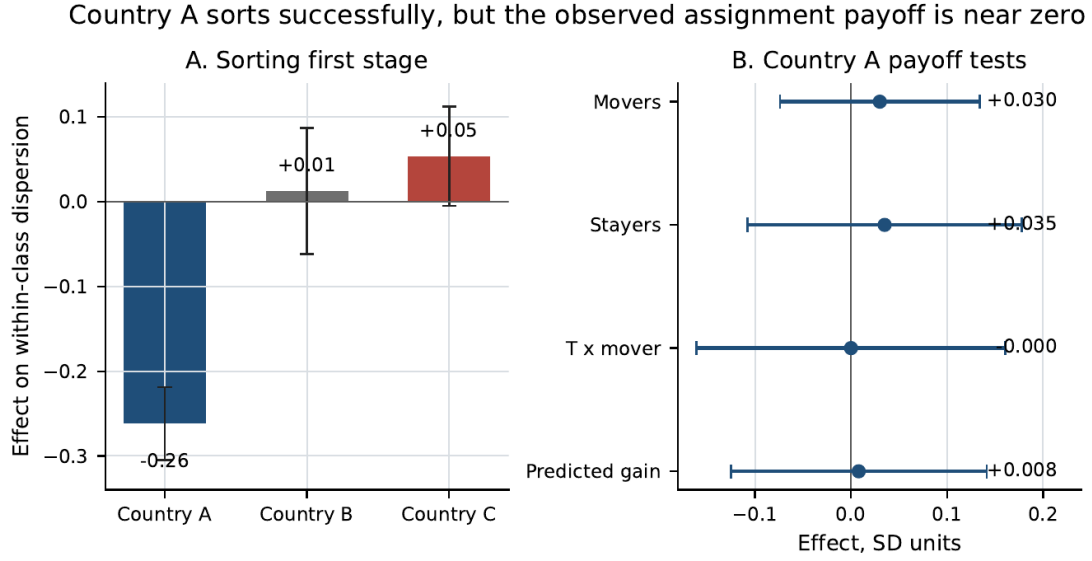
Figures

Figure 1: Three Experiments in Implementation Space



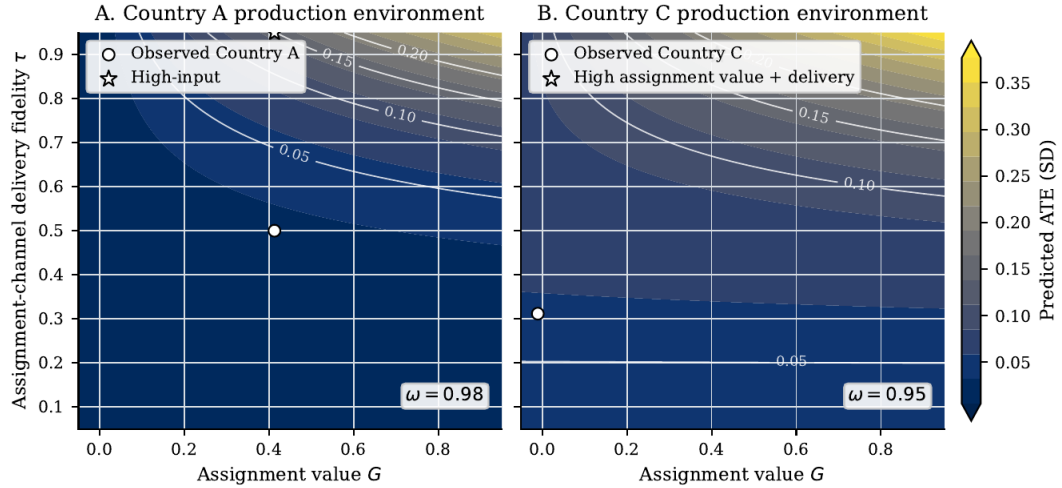
Notes: Panel A reports signal quality, assignment execution, and delivery fidelity. Panel B reports reduced-form ITT estimates with 95 percent confidence intervals. Country labels are anonymized.

Figure 2: First-Stage Sorting and Revealed Assignment Payoffs



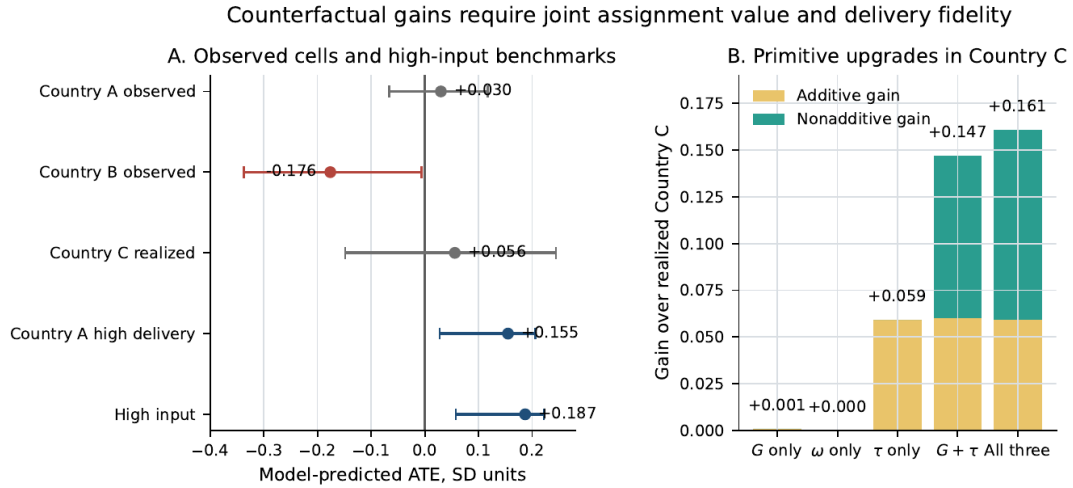
Notes: Panel A reports treatment effects on within-class baseline-score dispersion. Panel B reports Country A assignment-payoff diagnostics. Error bars are 95 percent confidence intervals.

Figure 3: Structural Complementarity Between Signal Quality and Delivery Fidelity



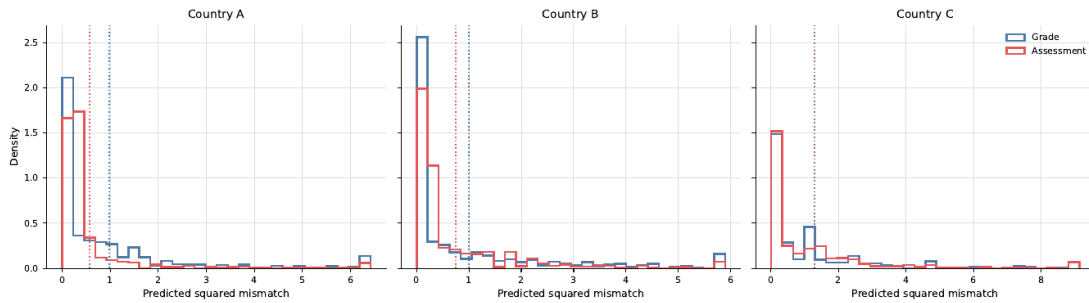
Notes: The figure plots model-predicted ATEs over assignment value and assignment-channel delivery fidelity. The surface illustrates how returns to measurement depend on delivery.

Figure 4: Structural Counterfactuals and Complementarity



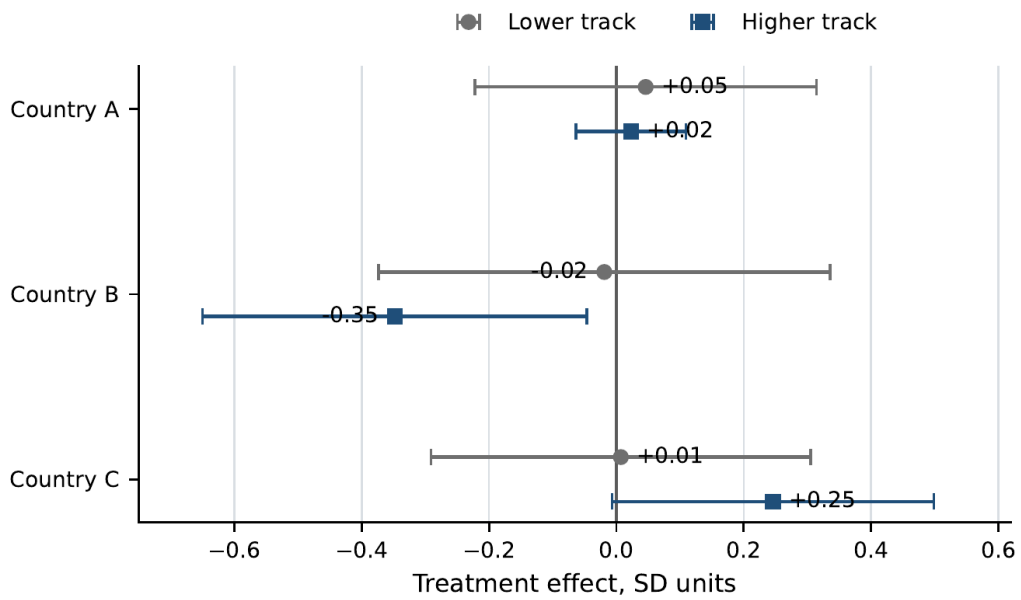
Notes: Panel A reports model-predicted ATEs and intervals. Panel B decomposes gains over realized Country C into additive and nonadditive primitive upgrades.

Figure 5: Predicted Mismatch Under Grade and Diagnostic Assignment



Notes: The figure compares predicted squared mismatch under grade-based assignment and assessment-based assignment. Vertical dotted lines mark experiment-specific means.

Figure 6: Treatment Effects by Track Position Across Experiments



Notes: Points report experiment-specific treatment effects for lower-track and higher-track students, with 95 percent confidence intervals. Country labels are anonymized.

Tables

Table 1: Incumbent Grade Signal and Diagnostic Increment

	Country A	Country B	Country C
Grade-only predictive power	0.011	0.059	0.129
Grade + diagnostic predictive power	0.534	0.109	0.247
Incremental diagnostic value	0.523	0.050	0.118
Within-grade rank persistence	0.684	0.242	0.388

Notes: Entries are control-group diagnostics from regressions predicting raw follow-up scores. Grade-only predictive power is the R^2 from grade fixed effects alone; grade + diagnostic adds the baseline diagnostic score. Incremental diagnostic value is the difference between the two R^2 values. Country A and Country B use the common 2018 literacy design. Country C reports the primary T3 endterm outcome; Appendix Table 44 reports all Country C waves. Within-grade rank persistence is the grade-weighted Spearman correlation between baseline and follow-up ranks.

Table 2: Pre-Experiment Assessment of General Delivery Quality

	Country B	Country A	Country C
Lesson completion	Low	Medium	High
Teacher quality	Low	Medium	High
School audit pass rate	—	19%	—
Programme quality	Established	Established	Newly introduced

Notes: Rankings are from pre-experiment implementation records collected before the Country C experiment. Country C introduced a new numeracy curriculum in both treatment and control arms; the earlier experiments used established literacy programmes. The Country A school audit pass rate is reported from comparable pre-experiment implementation records. The Country C experiment was selected because the setting had the highest lesson completion and teacher quality among the study settings. This table reports the *ex ante* general delivery quality ranking. Assignment-channel delivery fidelity (τ)—the fidelity with which the curriculum carrying the assignment-rule difference is delivered—does not follow this ranking: the assignment-linked numeracy curriculum in Country C achieved only 26% lesson completion (Table 3), well below the established curriculum’s 69%. The structural estimates (Section 6) imply $\hat{\tau}_{\text{Country A}} = 0.50 > \hat{\tau}_{\text{Country B}} = 0.38 > \hat{\tau}_{\text{Country C}} = 0.31$.

Table 3: Country C Operational Outcomes by Treatment Status

	Treatment	Control	<i>p</i> -value
Teacher attendance (%)	93	92	0.68
Pupil attendance (%)	64.5	64.0	0.88
School manager attendance (%)	75	77	0.71
Pupil-teacher ratio (T2)	20.4	21.8	0.54
Pupil-teacher ratio (T3)	20.7	23.1	0.33
Assignment-linked numeracy completion (%)	26.2	26.1	0.98
Math lesson completion (%)	69.4	68.7	0.85

Notes: Treatment-control differences in administrative data. Assignment-linked numeracy is the newly introduced numeracy curriculum; math is the pre-existing mathematics curriculum. Lesson completion is the fraction of scheduled lessons recorded as delivered. *p*-values are from regressions with strata fixed effects and school-level clustering.

Table 4: Three-Experiment Design Summary

	Country B	Country A	Country C
Year	2018	2018	2021
Subject	Literacy	Literacy	Numeracy
Grades per sub-experiment (<i>G</i>)	2	2	3
Tracks (<i>K</i>)	2	2	3
<i>K</i> vs. <i>G</i>	<i>K</i> = <i>G</i>	<i>K</i> = <i>G</i>	<i>K</i> = <i>G</i>
Schools (T / C)	~45 / ~45	50 / 250	20 / 22
<i>Three primitives</i>			
Signal quality ($\hat{\rho}$)	0.055	0.533	0.121
Assignment execution ($\hat{\omega}$)	1.00	1.00	0.84
Assignment-channel fidelity ($\hat{\tau}$)	0.38	0.50	0.31
Ex ante delivery ranking	Low	Medium	High
<i>Sorting accuracy detail</i>			
Incremental R^2	0.03–0.07	0.51–0.56	0.11–0.22
Placement binding	Deterministic	Deterministic	Spearman = 0.17
Delivery process moment	+1.1pp	+2.4pp	26% DI completion

Notes: Country B runs two parallel sub-experiments (Grades 1–2 and Grades 3–4), each with $K = G = 2$. $\hat{\rho}$ is estimated from control-group baseline–endline R^2 ; $\hat{\omega}$ and $\hat{\tau}$ are from the structural model (Section 6), estimated from audit/roster and curriculum-specific delivery moments, respectively. “Ex ante delivery ranking” is the pre-experiment assessment based on general lesson completion and teacher quality. The delivery process moment is the treatment-control difference in lesson completion for Country A and Country B, and the balanced assignment-linked numeracy completion level for Country C, where both arms received the new curriculum. Incremental R^2 is the within-grade variance explained by adding baseline scores to a grade-FE-only model.

Table 5: Predicted Mismatch Under Grade and Diagnostic Assignment

	N	Grade	Diagnostic	Reduction	Lower	Higher	Same
Country A	9718	0.982	0.569	0.413	57.2%	42.8%	0.0%
Country B	7584	0.993	0.745	0.248	42.5%	57.5%	0.0%
Country C	1256	1.286	1.297	-0.011	44.8%	55.2%	0.0%

Notes: The table uses the control-trained assignment-gain objects described in the text. “Grade” is the mean squared distance between the baseline ability proxy and the grade-level instructional target. “Diagnostic” replaces the grade target with the intended diagnostic-assigned track target. “Reduction” is Grade minus Diagnostic, so positive values mean the diagnostic rule lowers predicted mismatch. Lower, Higher, and Same report the share of students whose predicted mismatch falls, rises, or is unchanged. Country C uses the designed assignment rule; realized-assignment failures enter the implementation diagnostics separately.

Table 6: Three-Channel Decomposition and (ρ, ω, τ) Summary

	Country B	Country A	Country C
<i>Position in (ρ, ω, τ) space</i>			
Signal quality ($\hat{\rho}$)	0.055	0.533	0.121
Assignment execution	1.00	1.00	0.84
($\hat{\omega}$)			
Assignment-channel $\hat{\tau}$	0.38	0.50	0.31
<i>Assignment and delivery evidence</i>			
Δ within-class SD	small	-0.328	+0.104
Delivery process	—	+2.4pp	26% DI completion
<i>moment</i>			
<i>Residual classroom-production diagnostic</i>			
$\hat{\zeta}$ (BH estimate)	-0.048	-0.134**	≈ 0
Δ peer mean	+0.041	+0.277	-0.242
Peer/rank contribution	-0.002	-0.037	+0.026
<i>Organizational costs</i>			
Δ class size	+10.0	-2.9	-3.0
Audit: groups as	—	—	$\leq 50\%$
designed			
Overall ITT	-0.212	+0.031	+0.075
Remainder (ITT - peer/rank)	-0.210	+0.067	+0.049
Main interpretation	Weak signal	Payoff absent at observed fidelity	Execution and delivery failure

Notes: Each column summarizes one experiment’s position on the evidence used in the reduced-form interpretation. $\hat{\rho}$, $\hat{\omega}$, $\hat{\tau}$ are from the structural model (Section 6). Peer/rank contribution = $\hat{\zeta} \times \Delta\theta$. Remainder = ITT - peer/rank contribution in the channel-summary calculation. Country C ITT is pooled across waves; peer contribution uses T3 ETE estimates. The Country C delivery process moment is the balanced level of assignment-linked numeracy completion in treatment and control. The Country C class-size row reports classroom-level means from Table 47; the student-weighted T3 channel-summary sample has a different class-size contrast.

Table 7: Stage-4 Moment Blocks and Inputs

Block	Moments / inputs	Criterion weight	Role in estimation
Observed treatment effects	3 setting ITTs	inverse variance (54.9–356.0)	Fit ATEs while residuals are shrinkage-regularized.
Social-channel diagnostics	3 peer/rank composites	inverse diagnostic variance (100.0–200.0)	Discipline the common peer/rank scale.
Sorting first stage	3 within-class dispersion changes	scale weight (1000.0)	Separate sorting compression from assignment payoff.
Revealed assignment payoff	Country A treatment $\times$ predicted assignment-gain slope	inverse variance (216.3)	Pin down delivery activation at realized fidelity.
Fixed primitives and environment	$(\hat{\rho}, \hat{\omega}, \hat{\tau})$; peer shifts, class size, grade dispersion	not criterion targets	Measured inputs for observed and counterfactual regimes.

Notes: The stage-4 minimum-distance criterion targets only moments with a criterion weight in this table. Classroom peer shifts, class size, grade dispersion, and the first-three-block primitives enter the production mapping as measured inputs, not as additional fitted moments. This distinction prevents the target list from overstating what the outcome block estimates.

Table 8: Primitive Identification Moments

Primitive	Moment	Data	Model / estimate
<i>Block 1: Signal quality from control-group measurement moments</i>			
ρ	Country A: baseline–endline R^2 (control, $N = 4,195$)	0.533	$\hat{\rho} = 0.533$
ρ	Country B: baseline–endline R^2 (control, $N = 1,482$)	0.055	$\hat{\rho} = 0.055$
ρ	Country C: baseline–endline R^2 (control, $N = 413$)	0.121	$\hat{\rho} = 0.121$
<i>Block 2: Assignment execution from audit/roster moments</i>			
ω	Country A: cutoff misclassification rate	0.000	$\hat{\omega} = 1.00$
ω	Country B: cutoff misclassification rate	0.000	$\hat{\omega} = 1.00$
ω	Country C: score–group rank correlation	0.249	0.253
ω	Country C: share schools with ≤ 2 groups	0.200	0.203
ω	Country C: share schools with any upper track	0.800	0.789
ω	Country C: share missing assignment	0.348	0.340
<i>Block 3: Delivery fidelity from assignment-channel process moments</i>			
τ	Country A: lesson-completion treatment–control difference	0.025	$\hat{\tau} = 0.50$
τ	Country B: lesson-completion treatment–control difference	0.011	$\hat{\tau} = 0.38$
τ	Country C: assignment-linked numeracy completion, treatment	0.262	$\hat{\tau} = 0.31$
τ	Country C: assignment-linked numeracy completion, control	0.261	$\hat{\tau} = 0.31$

Notes: The table reports the primitive-specific moments used to pin down the first three structural primitives before the treatment-effect fit. Signal quality is the setting-level control-group baseline–endline R^2 . Country A and Country B assignment execution is deterministic cutoff compliance; Country C execution is fit to the cleaned roster/audit moments shown here, with conflicting external audit targets reserved for validation. Delivery fidelity is based on lesson-completion moments in the assignment channel; Country C uses assignment-linked numeracy completion because that curriculum carries the assignment-rule treatment.

Table 9: Delivery-Fidelity Calibration

Setting	Process evidence	Value	Calibration rule	Prior	$\hat{\tau}$	Role
Country A	Lesson-completion treatment-control difference	0.025	$\text{clip}(0.30 + 8\Delta LC, 0.05, 0.95)$	0.500	0.500	Estimation
Country B	Lesson-completion treatment-control difference	0.011	$\text{clip}(0.30 + 8\Delta LC, 0.05, 0.95)$	0.385	0.385	Estimation
Country C	Assignment-linked numeracy completion in treatment and control	T 0.262; C 0.261	$\text{clip}(0.10 + 0.80LC_T + 2(LC_T - LC_C), 0.05, 0.85)$	0.312	0.312	Estimation
Country C	Wrong-track / partial-script proxy	0.350	Not used in calibration	–	–	Diagnostic only

Notes: The table documents the outcome-free mapping from assignment-channel process evidence into the delivery-fidelity primitive. ΔLC is the treatment-control lesson-completion difference. LC_T and LC_C are assignment-linked numeracy completion rates in Country C treatment and control schools. The wrong-track proxy is retained as diagnostic evidence and is not used to set τ .

Table 10: Raw-Unit Interpretation of Delivery-Fidelity Benchmarks

Benchmark	τ	A/B ΔLC	Country C linked completion	Interpretation
Country A observed	0.500	2.5%	50.0%	Observed Country A index
Country B observed	0.385	1.1%	35.6%	Observed Country B index
Country C observed	0.312	0.1%	26.4%	Observed Country C index
Moderate delivery	0.700	5.0%	75.0%	Moderate benchmark
High delivery	0.900	7.5%	above Country C cap	High-delivery benchmark
High-input benchmark	0.950	8.1%	above Country C cap	High-input benchmark

Notes: The table translates selected values of the normalized delivery-fidelity index back into the raw process units used in Table 9. For Country A and Country B, the raw unit is the treatment-control lesson-completion difference implied by $\tau = 0.30 + 8\Delta LC$. For Country C, the raw unit is the assignment-linked numeracy completion rate implied by $\tau = 0.10 + 0.80LC$ when treatment and control completion are equal. The Country C calibration is capped at $\tau = 0.85$, so the $\tau = 0.90$ and $\tau = 0.95$ benchmarks should be read as high-delivery activation benchmarks rather than observed Country C-style completion rates.

Table 11: Sensitivity to Delivery-Fidelity Calibration

Spec.	τ_K	τ_L	τ_N	Max ITT/SE	K high- τ	High-input
Lower process map	0.40	0.31	0.22	0.297	+0.177	+0.205
Preferred	0.50	0.38	0.31	0.282	+0.152	+0.186
Upper process map	0.60	0.46	0.40	0.277	+0.124	+0.164

Notes: Each row re-estimates the stage-4 production block after replacing the stage-3 delivery-fidelity primitives with a transparent lower, preferred, or upper process-to- τ mapping. The lower and upper mappings preserve the same process moments and setting ordering logic but vary the intercept and slope used to translate completion evidence for the assignment channel into τ . The high-input counterfactual sets $G = \max_m \hat{G}_m$, $\omega = 0.98$, and $\tau = 0.95$; this table asks whether the observed-cell calibration of τ materially changes the extrapolation.

Table 12: Structural Parameter Estimates

	Country B	Country A	Country C
<i>Block 1: Signal quality (control-group moments)</i>			
$\hat{\rho}$	0.055	0.533	0.121
<i>Block 2: Assignment execution (audit/roster moments)</i>			
$\hat{\omega}$	1.00	1.00	0.84
<i>Block 3: Assignment-channel delivery fidelity (process moments)</i>			
$\hat{\tau}$	0.38	0.50	0.31
<i>Block 4: Production parameters (common across settings)</i>			
$\hat{\lambda}$ (assignment-payoff scale)		0.400	
$\hat{\varphi}$ (social channel)		0.192	
$\hat{\omega}_r$ (rank weight in composite)		-0.276	
$\hat{\chi}_N$ (class-size pressure)		0.033	
$\hat{\chi}_V$ (grade dispersion)		0.022	
$\hat{\alpha}$ (delivery activation)		4.615	
$\hat{\kappa}$ (sorting compression)		1.649	

Notes: Blocks 1–3 measure or calibrate primitives sequentially before fitting treatment-effect moments: control predictive content for ρ , implementation evidence for ω , and process/delivery evidence for τ . Block 4 is estimated by minimum distance conditional on fixed $(\hat{\rho}, \hat{\omega}, \hat{\tau})$, targeting harmonized ITTs, peer/rank composites, within-class dispersion changes, and the Country A assignment-payoff slope. λ , α , κ , φ , ω_r , χ_N , and χ_V are constrained to be common across settings; δ_m is setting-specific and shrinkage-regularized.

Table 13: Targeted Moments: Data vs. Model

Setting	Moment	Data	Model	Std. resid.	Obj.
Country A	ITT	+0.031	+0.030	+0.023	0.001
Country A	$\hat{\zeta}$ (peer/rank)	-0.134	-0.091	-0.615	0.378
Country A	Δ within-class SD	-0.193	-0.193	-0.000	0.000
Country A	assignment-payoff slope	+0.008	+0.009	-0.010	0.000
Country B	ITT	-0.212	-0.176	-0.267	0.072
Country B	$\hat{\zeta}$ (peer/rank)	-0.048	-0.111	+0.634	0.401
Country B	Δ within-class SD	+0.011	+0.011	-0.003	0.000
Country C	ITT	+0.093	+0.056	+0.282	0.079
Country C	$\hat{\zeta}$ (peer/rank)	-0.109	-0.125	+0.156	0.024
Country C	Δ within-class SD	+0.021	+0.021	-0.001	0.000

Notes: “Data” reports the reduced-form point estimate. “Model” reports the model-implied moment at the estimated parameter values. “Std. resid.” is $(\text{Data} - \text{Model})/\sqrt{w}$ and “Obj.” is the moment’s contribution to the stage-4 criterion. ITT targets are the harmonized study-specific estimates from Table 23. The assignment-payoff slope is the coefficient on treatment interacted with the standardized predicted assignment-gain index in Table 27. Because setting residuals are shrinkage-regularized, the model is not forced to interpolate the three ITTs exactly.

Table 14: Model Fit, Maintained Restrictions, and Sensitivity Checks

Item	Evidence	Status
Signal-quality ranking	$\rho_K = 0.533 > 0.121 > 0.055 = \rho_L$	Pass
Assignment-execution ranking	$\omega_K = \omega_L = 1.00 > 0.84 = \omega_N$	Pass
Delivery primitive measured before outcomes	τ calibrated from process moments before stage 4	Pass
Maintained restriction: Country A delivery	ATE rises from +0.030 to +0.152	Pass
Signed assignment value: Country C execution	$G_N = -0.014$; cleaner execution need not raise ATE	Pass
Country C target coherence	Cleaned roster moments estimated; conflicting audit targets diagnostic only	Pass
Observed-cell fit	Predicted observed ATEs: K +0.030, L -0.176, N +0.056; max ITT error/SE 0.282	Pass

Notes: The table separates measured-input checks, observed-cell fit, and maintained comparative-static restrictions. The comparative-static rows are restrictions imposed before interpreting counterfactuals; they are not validation evidence for the high-input result. The high-input ATE itself is reported only as a counterfactual result in Table 15.

Table 15: Counterfactual Treatment Effects

Scenario	ATE	90% interval	Description
<i>Observed</i>			
Country A observed	+0.030	[−0.066, +0.117]	$G = 0.53, \omega = 1.00, \tau = 0.50$
Country B observed	−0.176	[−0.337, −0.004]	$G = 0.32, \omega = 1.00, \tau = 0.38$
Country C realized	+0.056	[−0.149, +0.243]	$G = −0.01, \omega = 0.84, \tau = 0.31$
<i>Counterfactual 1: Country A with higher τ</i>			
Country A high- τ	+0.152	[+0.028, +0.210]	$G = 0.53, \omega = 1.00, \tau = 0.90$
<i>Counterfactual 2: Country C as designed</i>			
Country C designed ω	+0.056	[−0.149, +0.243]	$G = −0.01, \omega = 0.95, \tau = 0.31$
<i>Counterfactual 3: Fully idealized</i>			
High assignment value, high ω , high τ	+0.186	[+0.060, +0.223]	$G = 0.53, \omega = 0.98, \tau = 0.95$
<i>One-at-a-time decomposition (from Country C baseline)</i>			
Upgrade assignment value only	+0.057	[−0.141, +0.265]	$G \rightarrow 0.53, \omega = 0.84, \tau = 0.31$
Upgrade τ only	+0.086	[−0.132, +0.289]	$G = −0.01, \omega = 0.84, \tau = 0.90$
Upgrade ω only	+0.056	[−0.149, +0.243]	$G = −0.01, \omega = 0.95, \tau = 0.31$

Notes: G is the signed assignment-value primitive: mean predicted mismatch under grade assignment minus mean predicted mismatch under diagnostic assignment, normalized so Country A equals the highest signal-quality value. Signal quality ρ is still used in the sorting-compression equation. Intervals are 5th–95th percentiles from 49 parametric bootstrap re-estimation draws of the targeted stage-4 reduced-form moments, conditional on the stage-1 to stage-3 primitive estimates. The high- τ benchmark is $\tau = 0.90$ for the Country A and Country C one-at-a-time counterfactuals. The fully idealized case uses $G = 0.53, \omega = 0.98$, and $\tau = 0.95$. Country C designed execution sets $\omega = 0.95$. The one-at-a-time decomposition upgrades each primitive separately from Country C’s realized values.

Table 16: Structural Decomposition of Counterfactual ATEs

Scenario	Assign.	Social	Class/grade	Residual	ATE excl. resid.	ATE
Country A observed	+0.009	−0.000	−0.005	+0.027	+0.003	+0.030
Country A high delivery	+0.131	−0.000	−0.005	+0.027	+0.126	+0.152
Country C realized	−0.000	−0.039	−0.040	+0.134	−0.078	+0.056
Country C high assignment value + delivery	+0.110	−0.006	−0.040	+0.134	+0.065	+0.199
Country C all three	+0.125	−0.006	−0.040	+0.134	+0.079	+0.214
Fully high-input cell	+0.165	−0.000	−0.005	+0.027	+0.159	+0.186

Notes: Assignment is $\hat{\lambda}G\omega\tau^{\hat{\alpha}}$, where G is the signed assignment-value primitive. Social is the peer/rank composite scaled by $(1-\tau)$. Class/grade combines the class-size pressure and within-class grade-dispersion terms. Residual is the shrinkage-regularized setting productivity shifter δ_m ; it is reported to make the accounting transparent, not interpreted as a mechanism. “ATE excl. resid.” sets $\delta_m = 0$ while holding all common production parameters and measured environment inputs fixed.

Table 17: Complementarity of Primitive Upgrades in Country C

Scenario	G	ω	τ	ATE	Gain	Nonadd.
Realized Country C	-0.014	0.84	0.31	+0.056	+0.000	+0.000
Upgrade G only	0.533	0.84	0.31	+0.057	+0.001	+0.000
Upgrade ω only	-0.014	0.95	0.31	+0.056	-0.000	+0.000
Upgrade τ only	-0.014	0.84	0.90	+0.086	+0.030	+0.000
Upgrade $G + \omega$	0.533	0.95	0.31	+0.057	+0.001	+0.000
Upgrade $G + \tau$	0.533	0.84	0.90	+0.199	+0.143	+0.112
Upgrade $\omega + \tau$	-0.014	0.95	0.90	+0.086	+0.030	-0.000
Upgrade all three	0.533	0.95	0.90	+0.214	+0.158	+0.127

Notes: Each row evaluates the Country C production environment while upgrading the indicated primitives from Country C’s realized values. High assignment value sets G to Country A’s estimate, high execution sets $\omega = 0.95$, and high delivery sets $\tau = 0.90$. “Gain” is relative to realized Country C, and “Nonadd.” is the gain beyond the sum of the corresponding one-at-a-time gains. Because every row uses the same shrinkage-regularized Country C residual, the gain and nonadditive columns are invariant to that residual. The comparison is designed to show complementarity among primitives, not to replace the fully idealized high-input counterfactual in Table 15.

Table 18: Marginal Return to Assignment Value by Delivery Fidelity

Scenario	ω	τ	Gain $G_N \rightarrow G_K$	MP of G	Cross-partial
Country C realized	0.84	0.31	+0.001	0.002	0.023
Country C high delivery	0.84	0.90	+0.113	0.207	1.061
Country C high delivery + execution	0.95	0.90	+0.128	0.234	1.198
Country A observed delivery	1.00	0.50	+0.009	0.016	0.151
Country A high delivery	1.00	0.90	+0.134	0.246	1.261
Fully high-input	0.98	0.95	+0.169	0.309	1.503

Notes: The table evaluates the assignment-payoff term $\hat{\lambda}G\omega\tau^{\hat{\alpha}}$. “Gain” is the ATE increase from raising assignment value from Country C’s estimate to Country A’s estimate, holding the row’s ω and τ fixed. The last two columns report the analytic marginal product of assignment value and the assignment-value–delivery cross-partial implied by the preferred production map.

Table 19: Delivery-Fidelity Thresholds Under High Assignment Value and Execution

Production env.	Residual	$\tau_{\geq 0}$	$\tau_{\geq .10}$	$\tau_{\geq .18}$	ATE at $\tau = .95$
Country A	With δ_m	0.00	0.81	0.94	+0.186
Country A	$\delta_m = 0$	0.45	0.86	0.97	+0.159
Country C	With δ_m	0.00	0.64	0.85	+0.252
Country C	$\delta_m = 0$	0.75	0.93		+0.117

Notes: Each row holds assignment value at Country A’s estimate and assignment execution near clean, then varies delivery fidelity τ in the indicated production environment. The residual-included rows use the shrinkage-regularized setting residual; the residual-excluded rows set $\delta_m = 0$. Blank entries mean the target ATE is not reached for any $\tau \in [0, 1]$.

Table 20: Sensitivity to Auxiliary Scale Regularization

Specification	Raw norm	λ prior	φ prior	Max ITT/SE	High-input
Preferred	0.05	10.0	2.0	0.282	+0.186
No raw-norm penalty	0.00	10.0	2.0	0.293	+0.190
Weak scale priors	0.05	2.5	0.5	0.275	+0.179
No lambda/phi priors	0.05	0.0	0.0	0.284	+0.300
No auxiliary scale restriction	0.00	0.0	0.0	0.289	+0.368

Notes: Each row re-estimates the stage-4 production block holding the residual prior at its preferred value and varying only auxiliary scale regularization. “Raw norm” is the coefficient on the raw-parameter norm penalty; “ λ prior” and “ φ prior” are the weights on scale priors centered at 0.4 and 0.2, respectively. The high-input ATE sets assignment value G to Country A’s estimate, $\omega = 0.98$, and $\tau = 0.95$.

Table 21: Structural Benchmark Under Alternative Payoff Primitives

Payoff primitive	Country A	Country B	Country C	Max ITT/SE	K high- τ	High-input
Preferred assignment value	0.533	0.320	-0.014	0.282	+0.152	+0.186
Legacy rho payoff	0.533	0.055	0.121	0.281	+0.155	+0.187
Net mismatch reduction	0.533	0.320	0.000	0.282	+0.152	+0.186
Share-weighted reduction	0.533	0.238	0.000	0.282	+0.153	+0.186

Notes: The preferred model uses a signed assignment-value primitive in the assignment-payoff term: mean predicted mismatch under grade assignment minus mean predicted mismatch under diagnostic assignment, normalized so Country A equals the highest signal-quality value. Negative values are allowed, so a diagnostic rule that increases predicted mismatch lowers the assignment-payoff component. The “Legacy rho payoff” row re-estimates the old payoff specification using control-group predictive content directly. “Net mismatch reduction” truncates negative mean reductions at zero. “Share-weighted reduction” multiplies the positive mean reduction by the share of students whose predicted mismatch falls before normalization. The mechanical sorting equation continues to use ρ because it predicts classroom compression rather than payoff value. The table reports the Country A high-delivery and fully high-input counterfactuals under each production map.

Table 22: Pooled ITT Across Country A, Country B, and Country C

	(1)	(2)	(3)	(4)
	Country A	Country B	Country C	Pooled
Treatment	0.031 (0.053)	-0.212 (0.135)	0.093 (0.133)	-0.070 (0.071)
Control Mean	0.012	0.091	0.036	0.033
N	4954	3154	1661	9769
Pooled ITT (Country C T2 ETE alt)				-0.051 (0.067)

Notes: Outcome is standardized endline score for Country A/Country B and T3 ETE math+numeracy index for Country C. Controls: country-specific baseline slopes and country-nested strata FE. SEs clustered at country-specific cluster units (school for Country A, school-grade-group for Country B, school for Country C). Final row shows pooled robustness replacing Country C T3 ETE with T2 ETE index. ***, **, and * indicate 1%, 5%, and 10% significance.

Table 23: Pooled and Meta-Analytic Treatment Effects Across Experiments

Estimand or diagnostic	Estimate	SE / statistic
<i>Common-effect estimands</i>		
Student-weighted IPD	-0.070	(0.071)
Experiment-balanced IPD	-0.058	(0.077)
Fixed-effect meta-analysis	0.010	(0.046)
Random-effects meta-analysis	-0.008	(0.078)
<i>Study-specific effects from stacked IPD</i>		
Country A	0.031	(0.053)
Country B	-0.212	(0.135)
Country C	0.093	(0.131)
<i>Heterogeneity and primitive-score diagnostics</i>		
Treatment-effect heterogeneity test		p = 0.196
Meta-analysis Q statistic		Q = 3.270, p = 0.195
I-squared / tau-squared		38.8% / 0.007
Treatment $\times$ ($\rho\omega\tau$) slope	0.683	(0.494)
Implied effect: lowest/highest score	-0.132 / 0.035	descriptive
Student observations / studies	9769	3

Notes: IPD = individual participant data. Student-weighted IPD weights each student equally; experiment-balanced IPD weights each experiment equally. Meta-analytic rows pool the three study-specific estimates, with the random-effects SE using a Hartung-Knapp-style small-sample adjustment. All IPD specifications include country-specific baseline slopes and country-nested strata FE, with SEs clustered at harmonized assignment clusters. The primitive-score slope is descriptive because $\rho\omega\tau$ varies only at the experiment level.

Table 24: Treatment Effects by Track Position Across Countries

	(1)	(2)	(3)
	Country A	Country B	Country C
Lower-track effect	0.046 (0.137)	-0.019 (0.181)	0.007 (0.152)
Higher-track differential	-0.023 (0.147)	-0.328 (0.201)	0.239** (0.098)
Higher-track effect	0.023 (0.044)	-0.348** (0.154)	0.246* (0.129)
N	4954	3154	1611

Notes: Each column estimates a country-specific treatment-by-track-position interaction. The higher track is the upper reading group in Country A and Country B; in Country C it is the preferred two-group collapse, higher group (middle or upper intended track). Controls are country-specific baseline slopes and country-nested strata fixed effects. SEs are clustered at the harmonized assignment cluster.

Table 25: Country C Reduced-Form Effects Under the Preferred Two-Group Collapse

	T2 MTE	T2 ETE	T3 MTE	T3 ETE	Stacked ETE
<i>Effective two-track collapse: lower group versus higher group</i>					
Lower-group effect	-0.035 (0.139)	0.183* (0.098)	0.159 (0.147)	0.043 (0.155)	0.138 (0.102)
Higher-group effect	0.015 (0.089)	0.173* (0.101)	0.048 (0.168)	0.236* (0.132)	0.214** (0.095)
Higher group – lower group	0.050 (0.104)	-0.010 (0.103)	-0.111 (0.201)	0.194 (0.116)	0.076 (0.080)
N	1804	1947	1421	1611	3558

Notes: Outcomes are standardized math+numeracy index outcomes. Each column estimates a treatment-by-two-group interaction with baseline score, strata fixed effects, and grade fixed effects; the stacked endterm column pools T2 and T3 endterm observations and adds wave fixed effects. Standard errors are clustered at the school level. The table uses the preferred support-aware Country C collapse, combining the middle and upper intended tracks into a higher group because the realized upper-track cell is thin. ***, **, and * indicate 1%, 5%, and 10% significance.

Table 26: Pooled First Stage: Effect on Within-Class Dispersion

	(1)	(2)	(3)	(4)
	Country A	Country B	Country C	Pooled
Treatment	-0.262*** (0.022)	0.013 (0.038)	0.054* (0.030)	-0.071*** (0.023)
N	4954	3154	2354	10462

Notes: Harmonized dispersion measure is absolute deviation from classroom mean baseline ability. Country C uses `dev_b1`; Country A/Country B use `dev_eb`. Controls are country-specific baseline slopes and country-nested strata fixed effects. SEs are clustered at the harmonized assignment cluster.

Table 27: Country A: Assignment Payoff Diagnostics

Measure	Coef.	SE	N
<i>A. Randomized reallocation contrasts</i>			
Movers ITT	0.030	(0.053)	1816
Stayers ITT	0.035	(0.073)	3138
Treatment $\times$ mover	-0.000	(0.082)	4954
<i>B. Assignment intensity among movers</i>			
G1 movers up: Treatment $\times$ cutoff distance	0.005	(0.017)	1286
G2 movers down: Treatment $\times$ cutoff distance	-0.006	(0.023)	530
<i>C. Predicted assignment gain</i>			
Treatment $\times$ predicted assignment gain (1 SD)	0.008	(0.068)	4954
Top predicted-gain tercile ITT	-0.004	(0.056)	1547
Treatment $\times$ top predicted-gain tercile	-0.036	(0.089)	4954
<i>D. Descriptive control-group revealed-value checks</i>			
Mismatch proxy	-0.008	(0.011)	4195
Mismatch proxy + school-grade mean BL	-0.002	(0.011)	4195

Notes: Outcome is the standardized endline test score. Movers are students whose treatment reading-group assignment differs from their enrolled grade: Grade 1 students above the upper-track cutoff and Grade 2 students at or below the lower-track cutoff. Cutoff distance is signed so larger values indicate more clearly assigned movers. Predicted assignment gain is the standardized reduction in squared EB-score mismatch from replacing the grade target with the treatment-track target; the regression also controls for its square. Top predicted-gain tercile is defined over the same predicted assignment-gain measure. Panel D is estimated only in control classrooms and is descriptive rather than randomized. Standard errors are clustered at the school level.

Table 28: Summary Statistics — Country B

Variable	Experiment (Country B)			
	(1)	(2)	(3)	(4)
	Control Mean	Treatment Mean	SD (All)	N
Baseline score	21.014	19.848	7.936	4784
Endline score	25.983	24.467	7.612	3154
Upper-track assignment	0.612	0.549	0.494	4784
Has endline	0.659	0.660	0.474	4784
Class size	45.596	59.278	27.288	4784

Notes: Means and standard deviations are reported in the analytic sample. Endline outcomes are based on observations with non-missing endline test scores.

Table 29: Summary Statistics — Country A

Variable	Experiment (Country A)			
	(1)	(2)	(3)	(4)
	Control Mean	Treatment Mean	SD (All)	N
Baseline score	39.207	39.904	8.786	7102
Endline score	39.333	40.258	8.519	4954
Upper-track assignment	0.614	0.652	0.486	7102
Has endline	0.686	0.771	0.459	7102
Class size	24.307	20.935	11.233	7102

Notes: Means and standard deviations are reported in the analytic sample. Endline outcomes are based on observations with non-missing endline test scores.

Table 30: Country C Summary Statistics

Variable	Control Mean	Treatment Mean	SD (All)	N
Baseline score	82.608	80.171	17.719	2354
Observed T2 midterm	0.634	0.622	0.483	3737
Observed T2 endterm	0.677	0.691	0.465	3737
Observed T3 midterm	0.516	0.470	0.500	3737
Observed T3 endterm	0.510	0.650	0.494	3737
Lower-group share	0.356	0.609	0.499	2441
Higher-group share	0.644	0.391	0.499	2441
Number of schools	22	20	—	42

Notes: Means and SDs at student level. Follow-up indicators equal 1 if observed in that wave.

Table 31: Baseline Balance

	Experiment 1 (Country A)			Experiment 2 (Country B)		
	Control#mean	Coef.	N	Control#mean	Coef.	N
	(1)	(2)	(3)	(4)	(5)	(6)
Class size	19.094	-3.949*** (1.261)	213	22.357	2.652 (2.497)	99
Lower grade class size	19.697	-3.563** (1.387)	212	23.286	1.377 (2.630)	99
Upper grade class size	18.528	-4.370*** (1.292)	213	21.224	3.158 (2.830)	99
Baseline test score	0.037	0.040 (0.074)	213	-0.061	-0.150 (0.118)	99
Lower grade baseline score	0.010	0.096 (0.098)	212	0.030	-0.127 (0.155)	99
Upper grade baseline score	0.066	-0.018 (0.075)	213	-0.120	-0.198 (0.128)	99

Notes: Country A specifications are at the school level and control for strata fixed effects, with standard errors clustered at the school level. Country B specifications are at the school grade group level and control for randomization strata fixed effects, with standard errors clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 32: Country C Baseline Balance

Variable	Treat mean	Control mean	Diff	SE	p value
bl_score	80.171	82.608	-3.009	(2.570)	0.248
has_bl	0.606	0.651	-0.023	(0.033)	0.477
has_t2_mte	0.622	0.634	-0.027	(0.041)	0.521
has_t2_ete	0.691	0.677	-0.001	(0.034)	0.971
has_t3_mte	0.470	0.516	-0.103	(0.073)	0.163
has_t3_ete	0.650	0.510	0.135	(0.084)	0.117

Notes: Follow-up indicators are 1 if student observed in that wave. Diff column uses OLS with strata and grade FE, clustered by school.

Table 33: Test Score Follow-up

	Experiment 1 (Country A)			Experiment 2 (Country B)		
	Control#mean (1)	Coef. (2)	N (3)	Control#mean (4)	Coef. (5)	N (6)
Full sample	0.674	0.047 (0.036)	6,206	0.585	0.007 (0.060)	4,784
Lower grade	0.690	0.050 (0.043)	3,193	0.560	0.045 (0.073)	2,434
Upper grade	0.657	0.035 (0.056)	3,013	0.613	-0.029 (0.079)	2,350

Notes: All specifications use controls for baseline score, grade dummies (where applicable) and randomization strata fixed effects. Standard errors are clustered at the school-grade-group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 34: Country C Attrition by Term and Window

Wave/Outcome	Treat retain	Control retain	Diff	SE	p value
t2_mte_math	0.608	0.628	-0.027	(0.042)	0.517
t2_mte_num	0.610	0.589	0.020	(0.061)	0.747
t2_ete_math	0.687	0.676	-0.004	(0.034)	0.902
t2_ete_num	0.691	0.676	-0.001	(0.034)	0.985
t3_mte_math	0.460	0.514	-0.116	(0.075)	0.128
t3_mte_num	0.464	0.504	-0.100	(0.074)	0.183
t3_ete_math	0.650	0.510	0.135	(0.084)	0.117
t3_ete_num	0.636	0.493	0.143	(0.089)	0.118

Notes: Retention indicators defined unconditionally for all students in sample. OLS with strata and grade FE, clustered by school.

Table 35: Sample Flow — Country B

Sample flow step	Experiment (Country B)		
	(1)	(2)	(3)
	Control	Treatment	Total
All observations	3420	4165	7585
Analytic sample	2249	2535	4784
With baseline score	2249	2535	4784
With endline score	1482	1672	3154

Notes: Counts shown by treatment status at each sample construction step in the cleaned analysis dataset. The analytic sample keeps students with a nonmissing baseline score and at least two baseline scored students in the assignment unit used to form peer and classroom measures.

Table 36: Sample Flow — Country A

Sample flow step	Experiment (Country A)		
	(1)	(2)	(3)
	Control	Treatment	Total
All observations	8271	1447	9718
Analytic sample	6117	985	7102
With baseline score	6117	985	7102
With endline score	4195	759	4954

Notes: Counts shown by treatment status at each sample construction step in the cleaned analysis dataset. The analytic sample keeps students with a nonmissing baseline score and at least two baseline scored students in the assignment unit used to form peer and classroom measures.

Table 37: Country C Sample Flow

Step	Control	Treatment	Total
All observations	1995	1742	3737
Non-missing treat and school	1995	1742	3737
With baseline score	1299	1055	2354
Observed in T2 midterm	1265	1084	2349
Observed in T2 endterm	1350	1203	2553
Observed in T3 midterm	1029	819	1848
Observed in T3 endterm	1018	1133	2151

Notes: Sample-flow counts in the Country C cleaned wide analysis file.

Table 38: Effect on T2 Endterm Outcomes

	(1)	(2)	(3)
	Math	Numeracy	Math + Numeracy Index
Treatment	0.151 (0.115)	0.135 (0.083)	0.146* (0.082)
Control Mean	-0.000	0.000	-0.000
N	2032	2037	2038

Notes: OLS with baseline score control, strata FE, and grade FE. SEs clustered at school level. ***, **, and * indicate 1%, 5%, and 10%.

Table 39: Effect on T3 Endterm Outcomes

	(1)	(2)	(3)
	Math	Numeracy	Math + Numeracy Index
Treatment	0.040 (0.150)	0.104 (0.144)	0.105 (0.132)
Control Mean	-0.000	0.000	-0.015
N	1661	1616	1661

Notes: OLS with baseline score control, strata FE, and grade FE. SEs clustered at school level. ***, **, and * indicate 1%, 5%, and 10%.

Table 40: Effects Using Data Over Terms (Stacked)

	(1)	(2)	(3)
	Math	Numeracy	Math + Numeracy Index
Treatment	0.075 (0.090)	0.080 (0.081)	0.075 (0.076)
Control Mean	-0.000	0.000	0.002
N (student-waves)	7010	6905	7060

Notes: Pooled OLS on stacked student-wave observations with wave FE, baseline score control, strata FE, and grade FE. SEs clustered at school level. ***, **, and * indicate 1%, 5%, and 10%.

Table 41: Over-Term Effects by Track Position (lower vs. higher groups, Stacked)

	Math	Numeracy	Math + Numeracy Index
Lower-group effect	0.016 (0.104)	0.156 (0.104)	0.077 (0.091)
Middle-group effect	0.035 (0.121)	0.049 (0.116)	0.037 (0.109)
Upper-group effect	0.306* (0.173)	0.127 (0.124)	0.225 (0.141)
N (student-waves)	6741	6632	6783

Notes: Pooled OLS on stacked student-wave observations with wave FE. Rows report track-specific total treatment effects. the lower group is the treatment coefficient; the middle and upper groups are Treatment plus the corresponding interactions. SEs clustered at school level. Controls as in baseline tables.

Table 42: Baseline-Endline Predictive Power

Study	Control-group diagnostics			
	(1)	(2)	(3)	(4)
	Grade	N (Control)	Corr.	R^2
Country A	1	2190	0.713	0.509
	2	2005	0.748	0.559
Country B	1	411	0.247	0.061
	2	399	0.240	0.057
	3	324	0.184	0.034
	4	348	0.255	0.065

Notes: Correlation is the Pearson correlation between baseline and endline raw scores in the control group, by grade. R^2 is the squared correlation and measures the fraction of endline variance predicted by the baseline diagnostic.

Table 43: Alternative Signal-Quality Diagnostics

	Experiments	
	(1)	(2)
	Country A	Country B
R-squared grade FE only	0.011	0.059
R-squared grade FE + baseline	0.534	0.109
Incremental R-squared	0.523	0.050
Within-grade rank persistence	0.684	0.242

Notes: All computed in control group. Incremental R-squared = additional variance explained by adding baseline score to a grade FE only model. Rank persistence is the within grade Spearman correlation between baseline and endline score ranks, weighted by grade size.

Table 44: Country C Signal Quality (Incremental R-squared, Control Group)

	T2 MTE	T2 ETE	T3 MTE	T3 ETE
R-squared grade FE only	0.023	0.083	0.224	0.129
R-squared grade FE + baseline	0.238	0.299	0.332	0.247
Incremental R-squared	0.215	0.216	0.108	0.118
Within-grade rank persistence	0.471	0.489	0.377	0.388

Notes: Control group only. Incremental R-squared = variance explained by adding baseline score to a grade FE only model predicting the raw follow-up index (math+numeracy average). Rank persistence is the within grade Spearman correlation, weighted by grade size. Comparable to Country A (0.523) and Country B (0.050).

Table 45: Deterministic Assignment Cutoffs

	Experiments					
	(1)	(2)	(3)	(4)	(5)	(6)
	Country A			Country B		
	Cutoff	N treated	Rule error	Cutoff	N treated	Rule error
	Grade					
1	40	527	0.000	23	637	0.000
2	35	458	0.000	23	662	0.000
3				14	621	0.000
4				14	615	0.000

Notes: The table reports score thresholds used to assign treated students to the upper reading track. Rule error is the share of treated students whose observed upper-track assignment differs from the published deterministic cutoff rule.

Table 46: Classroom Reallocation under Cross-Grade Grouping

	Experiments			
	(1)	(2)	(3)	(4)
	Country A		Country B	
	Control	Treat	Control	Treat
Class size	20.88	17.99	32.19	42.16
Within-class grade variance	0.000	0.209	0.000	0.211
Number of grades in class	1.000	1.896	1.000	1.981
Squared track target misfit	1.352	0.419	1.519	0.639
Peer mean EB ability	-0.006	0.254	0.009	0.032

Notes: Entries are arm-specific means in the structural analysis sample. Realized classroom margins use treated-track classes in treatment and grade classes in control. Squared track target misfit is the square of EB ability minus the track target.

Table 47: Country C: Classroom Reallocation under Ability Grouping

	Control	Treatment
Class size	30.2	27.2
Within class grade variance	0.000	0.445
Number of grades in class	1.000	2.609
Within class baseline SD	0.686	0.809
Peer mean baseline ability	0.098	-0.109

Notes: Arm-specific means at the classroom level. Treatment classrooms are school-by-group; control classrooms are school-by-grade.

Table 48: Country C: Effect on Within-Class Dispersion

	(1) Baseline	(2) T2 ETE	(3) T3 ETE
Treatment	0.104* (0.056)	-0.008 (0.039)	0.005 (0.029)
Control Mean	0.547	0.452	0.411
N	2354	2038	1661

Notes: Outcome is absolute deviation of student score from classroom mean. Classroom = school $\times$ group (treatment) or school $\times$ grade (control). Baseline dispersion column omits baseline control. All columns include constituency and grade FE. SEs clustered at school. ***, **, and * indicate 1%, 5%, and 10%.

Table 49: Treatment Effects by Track and Ability Tercile (Country B)

Track	Ability Bin	Country B			
		(1)	(2)	(3)	(4)
		ITT	SE	p value	N
Lower	Bottom	-0.192	(0.214)	0.372	460
	Middle	0.150	(0.283)	0.599	459
	Top	-0.055	(0.238)	0.819	393
Upper	Bottom	-0.204	(0.174)	0.244	626
	Middle	-0.472*	(0.242)	0.055	606
	Top	-0.354**	(0.150)	0.021	610

Notes: Within-track ability terciles. OLS with strata FE and EB control. Standard errors are clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 50: Country C: Treatment Effects by Two-Group Track Position and Within-Group Ability Tercile (T3 ETE)

Group	Ability Bin	ITT	SE	p value	N
Lower group	Bottom	-0.229	(0.186)	0.228	249
	Middle	0.134	(0.208)	0.523	171
	Top	0.196	(0.190)	0.311	278
Higher group	Bottom	0.126	(0.199)	0.529	320
	Middle	0.168	(0.148)	0.263	354
	Top	0.355**	(0.150)	0.025	239

Notes: Country C is grouped using the preferred two-group collapse: lower group versus higher group. Within-group ability terciles are based on baseline score. OLS with baseline score, strata FE, grade FE, SEs clustered at school. ***, **, and * indicate 1%, 5%, and 10%.

Table 51: ITT Heterogeneity by Distance from Assignment Cutoff (Country B)

Quintile	Country B			
	(1)	(2)	(3)	(4)
	Mean $ s - c $	ITT	SE	N
Q1	1.7	-0.128	(0.174)	813
Q2	5.0	-0.064	(0.157)	636
Q3	8.0	-0.188	(0.148)	556
Q4	11.3	-0.383*	(0.198)	604
Q5	17.2	-0.385*	(0.219)	545
$T \times s - c $		-0.0253	(0.0176)	3154

Notes: Quintiles of absolute distance from the grade group assignment cutoff. OLS with strata FE and EB ability control. Standard errors are clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 52: Effect on Within-Class Dispersion in Test Scores

	Baseline Scores			Predicted Outcomes			Endline Scores		
	Control mean (1)	Coef. (2)	N (3)	Control mean (4)	Coef. (5)	N (6)	Control mean (7)	Coef. (8)	N (9)
Panel A: Country A									
Full Sample	0.663	-0.257*** (0.026)	7102	0.771	-0.267*** (0.021)	7102	0.661	-0.130*** (0.026)	4954
Lower Grade	0.648	-0.215*** (0.032)	3618	0.757	-0.222*** (0.027)	3618	0.659	-0.135*** (0.033)	2594
Upper Grade	0.678	-0.307*** (0.031)	3484	0.785	-0.321*** (0.027)	3484	0.663	-0.125*** (0.039)	2360
Panel B: Country B									
Full Sample	0.542	-0.022 (0.029)	4784	0.669	-0.042 (0.034)	4784	0.448	0.092** (0.042)	3154
Lower Grade	0.529	0.029 (0.035)	2417	0.648	0.034 (0.040)	2417	0.421	0.059 (0.063)	1679
Upper Grade	0.554	-0.074 (0.046)	2367	0.689	-0.115** (0.054)	2367	0.481	0.135** (0.053)	1475

Notes: The outcome is the absolute deviation of each student's standardized score from the classroom mean. All specifications include randomization-strata fixed effects. The baseline score columns omit the baseline control because the outcome is constructed from baseline scores; the predicted-outcome and endline score columns control for standardized baseline score. Standard errors are clustered at the school level in Country A and at the school grade group level in Country B. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 53: Treatment Effects on Lesson Completion Rate (Country A)

	Lesson completion rate			
	(1)	(2)	(3)	(4)
Treat	0.018 [0.014]	0.022 [0.015]	0.018 [0.016]	0.024 [0.017]
Teacher attendance		0.085*** [0.029]		0.090*** [0.034]
Class size			-0.001 [0.001]	-0.001 [0.001]
Control mean	0.172	0.172	0.173	0.173
Observations	1128	1122	780	776

Notes: Each column reports the coefficient from a regression of lesson completion rate on treatment status. The unit of observation is a school–subject–grade cell. Lesson completion is the fraction of scripted literacy-revision lessons completed, recorded in administrative records. Columns 2–4 add controls for teacher attendance rate and reading class size. Control means are computed in each column’s estimation sample. Standard errors are clustered at the school level in brackets. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 54: Approximate Peer/Rank Diagnostic (Borusyak–Hull Control Function)

	Experiments	
	(1)	(2)
	Country A	Country B
Peer mean EB ability	-0.133** (0.052)	-0.048 (0.108)
N	4954	3153

Notes: Dependent variable is standardized endline score. Peer mean EB ability is the mean of classmates’ EB ability, excluding the focal student. This is the approximate control function implementation: controls include baseline-decile $\times$ treatment $\times$ grade FE and strata FE. Standard errors are clustered at the school level in Country A and at the school grade group level in Country B. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 55: Country C: Peer Composition Effects (Borusyak-Hull)

	(1) T2 ETE Index	(2) T3 ETE Index
Leave-out peer mean	0.087 (0.066)	-0.109 (0.100)
N	2036	1659

Notes: Leave-out peer mean is the standardized mean baseline score of all other students in the same classroom (school $\times$ group for treated, school $\times$ grade for control), excluding the focal student. Controls: baseline-decile $\times$ treatment $\times$ grade FE, strata FE. SEs clustered at school. ***, **, and * indicate 1%, 5%, and 10%.

Table 56: Pooled Peer Composition Effects (BH-style)

	(1)	(2)	(3)	(4)
	Country A	Country B	Country C	Pooled
Peer baseline ability (leave-out mean)	-0.133** (0.052)	-0.048 (0.108)	-0.121 (0.098)	-0.105** (0.046)
N	4954	3153	1659	9766

Notes: Borusyak-Hull-style reduced form with baseline-decile $\times$ treatment $\times$ grade FE and country-nested strata FE. Peer variable is harmonized leave-self-out baseline ability (**peer_eb** for Country A/Country B, **peer_lo_b1** for Country C). SEs are clustered at the harmonized assignment cluster.

Table 57: Country A Peer/Rank Diagnostics: Approximate vs. Exact Borusyak–Hull Specifications

	(1)	(2)	(3)	(4)	(5)
	Decile FE	Ventile FE	Exact CF	Exact recentered	Exact CF (simple)
Peer mean EB ability	-0.133** (0.052)	-0.136** (0.053)	0.001 (0.085)	-0.009 (0.083)	-0.003 (0.075)
Exact μ_i^{BH} control			-0.177* (0.096)		-0.166* (0.088)
Own-score controls	Cell FE	Cell FE	Cubic EB $\times$ T $\times$ grade	Cubic EB $\times$ T $\times$ grade	Linear EB $\times$ T $\times$ grade
Strata FE	Yes	Yes	Yes	Yes	Yes
School-clustered SE	Yes	Yes	Yes	Yes	Yes
Observations	4954	4954	4952	4952	4952

Notes: Columns (1)–(2) report the approximate Country A peer/rank diagnostic specifications using baseline score bins $\times$ treatment $\times$ grade fixed effects. Column (3) is the exact Borusyak–Hull control function specification: it regresses standardized endline score on the realized peer mean, excluding the focal student, and the exact expected peer regressor $\mu_i^{BH} = p_j P_i(1) + (1 - p_j) P_i(0)$, where $P_i(1)$ is the peer mean under the deterministic treated reading-group rule and $P_i(0)$ is the peer mean under grade-based control classrooms. Column (4) uses the exact recentered regressor $\tilde{P}_i = P_i^{obs} - \mu_i^{BH}$. Column (5) keeps the exact control function but replaces the flexible cubic own-score controls with a simpler linear EB $\times$ treatment $\times$ grade control set. All exact specifications include strata FE and clustered standard errors at the school level. The probability p_j is the constituency-specific assignment probability from the full Country A year-1 randomization roster (50 of 292 schools assigned to treatment). ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 58: Peer/Rank Diagnostics and Accounting: Country A

	Country A specifications	
	(1)	(2)
	Decile FE	Ventile FE
<i>Panel A: Peer/rank composite ($\hat{\zeta}$)</i>		
Effect of 1 SD increase in peer mean	-0.133** (0.052)	-0.136** (0.053)
Observations	4954	4954
<i>Panel B: Treatment-induced classroom shifts</i>		
Δ mismatch, $ \hat{\theta} - \hat{I} $ (C – T)		0.421
Δ peer composition (T – C)		0.277
Δ class size (T – C)		-3.1
Δ within class baseline SD (C – T)		0.326
<i>Panel C: Accounting decomposition</i>		
Overall ITT		0.031 (0.052)
Peer/rank contribution ($\hat{\zeta} \times \Delta \bar{\theta}$)		-0.037 (0.019)
Remainder		0.067 (0.053)

Notes: Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient from regressing standardized endline scores on peer mean ability, excluding the focal student, conditioning on baseline score bin $\times$ treatment $\times$ grade fixed effects and strata FE. Standard errors are clustered at the school level. Panel B reports mean treatment control differences in classroom characteristics. Mismatch is measured as $|\hat{\theta}_i^{EB} - \hat{I}_k|$ where $\hat{I}_k$ is the population-level track mean. Panel C reports the implied peer/rank contribution under the approximate specification ($\hat{\zeta} \times$ observed peer shift) and a remainder that includes assignment gains, productive-time effects, and other non-peer/rank channels. Bootstrap SEs (1,000 replications, clustered at the school level) in parentheses in Panel C.

Table 59: Peer/Rank Diagnostics and Classroom Shifts: Country B

	Country B
	(1)
	Estimate
<i>Panel A: Approximate Borusyak–Hull peer/rank coefficient</i>	
$\hat{\zeta}_{Lib}$	-0.048 (0.108)
Observations	3153
<i>Panel B: Treatment-induced classroom shifts</i>	
Δ absolute EB-to-target mismatch (C – T)	0.348
Δ peer composition (T – C)	0.041
Δ class size (T – C)	11.4
Δ within class baseline SD (C – T)	0.058
Overall ITT	-0.211 (0.135)

Notes: Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient using decile $\times$ treatment $\times$ grade FE. Standard errors clustered at the school grade group level. Panel B reports treatment control differences in classroom characteristics. The near-zero peer/rank coefficient and small classroom composition shift provide little evidence that the measured peer/rank channel explains the negative Country B ITT pattern.

Table 60: Country C: Peer Effects and Accounting Decomposition (T3 ETE)

<i>Panel A: Peer/rank composite</i>	
Effect of 1 SD increase in peer mean	-0.109 (0.100)
<i>Panel B: Treatment-induced shifts</i>	
Δ within class dispersion (C – T)	-0.136
Δ peer composition (T – C)	-0.242
Δ class size (T – C, student-wtd.)	2.9
<i>Panel C: Accounting</i>	
Overall ITT	0.105 (0.132)
Peer/rank contribution ($\hat{\zeta} \times \Delta\bar{\theta}$)	0.026
Remainder	0.079

Notes: Panel A reports the BH peer estimate from regressing the T3 ETE index on leave-out peer mean, conditioning on baseline-decile $\times$ treatment $\times$ grade FE and strata FE, clustered at school. Panel B reports student-weighted mean treatment control differences in the T3 accounting sample. Classroom level means are reported separately in Table 47. Panel C decomposes the ITT into the peer channel and a remainder.

Table 61: Approximate Peer/Rank Diagnostic by Score Density

	Score-density tercile		
	(1) Low (tails)	(2) Medium	(3) High (middle)
<i>Panel A: Approximate peer/rank estimate by density tercile</i>			
Peer mean EB ability ($\hat{\zeta}$)	-0.179** (0.081)	-0.099* (0.053)	-0.123** (0.062)
N	1708	1672	1574
<i>Panel B: Interaction specification</i>			
Peer mean EB ability ($\hat{\zeta}$)	-0.141		
Peer EB $\times$ density interaction	-0.024 (0.039)		

Notes: Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient ($\hat{\zeta}$) separately for terciles of the baseline score density evaluated at each student’s own score. Students in the tails of the distribution (low density) have few classmates at similar ability levels, so their within class rank is insensitive to changes in peer composition. Students in the middle (high density) have many nearby classmates, so rank shifts more with peer changes. In the stylized rank model, $\hat{\zeta} \approx \pi + \eta \cdot (-f(\theta_i))$, where π is the mean peer effect, η is the rank effect, and $f(\theta_i)$ is the score density. If rank effects matter in this reduced-form composite ($\eta \neq 0$), $\hat{\zeta}$ would tend to be more negative in the high-density tercile, where rank is more responsive to peer shifts. Panel B reports a pooled interaction of peer mean with standardized density; the interaction is a diagnostic for whether the peer/rank coefficient changes with local score density. All specifications include baseline-decile $\times$ treatment $\times$ grade FE and strata FE, so they should be read as approximate cell-control specifications rather than exact design-based recentering checks. Density is estimated using a Gaussian kernel with Silverman bandwidth, computed within grade, and standardized to mean zero and unit variance within grade. Standard errors are clustered at the school level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 62: Country C: Peer Effect Heterogeneity by Score Density (T3 ETE)

	Score-density tercile		
	(1) Low (tails)	(2) Medium	(3) High (middle)
<i>Panel A: BH peer estimate by density tercile</i>			
Peer mean ($\hat{\zeta}$)	0.884** (0.351)	-0.173 (0.122)	-0.048 (0.113)
N	27	726	906
<i>Panel B: Interaction specification</i>			
Peer mean ($\hat{\zeta}$)		-0.205	
Peer $\times$ density ($\approx \hat{\eta}$)		0.134 (0.066)	

Notes: Panel A reports BH peer estimates by tercile of baseline score density. Panel B interacts peer mean with standardized density. All specifications include baseline-decile $\times$ treatment $\times$ grade FE and strata FE. SEs clustered at school. ***, **, and * indicate 1%, 5%, and 10%.

Table 63: ITT Estimates under Alternative Specifications

	Country A		Country B	
	Coef.	SE	Coef.	SE
Baseline (EB + strata FE)	0.031	(0.053)	-0.212	(0.135)
Raw baseline score control	0.031	(0.053)	-0.212	(0.135)
No baseline control	0.110	(0.071)	-0.240*	(0.139)
Wild cluster bootstrap p -value	0.622		0.126	
Permutation p -value	0.573		0.128	
N	4954		3154	

Notes: Each row reports the ITT estimate of treatment on standardized endline scores under a different specification. The baseline controls for empirical Bayes ability and strata fixed effects, with standard errors clustered at the school level in Country A and at the school grade group level in Country B. The raw baseline score row replaces the EB shrinkage estimate with the standardized raw baseline score. Wild cluster bootstrap p -values use 999 replications (Cameron et al. 2008). Permutation p -values are from a Fisher-exact randomization test with 999 permutations within strata at the country-specific assignment unit: school in Country A and school grade group in Country B. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 64: Reduced-Form Inference Checks

Experiment	ITT	Preferred SE	School SE	Wild p	RI p	N
Country A	0.031	(0.053)	(0.053)	0.622	0.573	4954
Country B	-0.212	(0.135)	(0.135)	0.126	0.128	3154
Country C	0.105	(0.132)	(0.132)	0.423	0.493	1661

Notes: The ITT is the primary country endpoint: Country A and Country B end-line literacy scores and Country C T3 endterm math–numeracy index. Preferred standard errors cluster at the randomization unit: school in Country A, school-grade group in Country B, and school in Country C. The school-SE column reclusters Country B at the school level; for Country A and Country C it repeats the preferred school cluster. Wild-cluster bootstrap p -values use 999 replications when the `boottest` package is available. RI p -values are Fisher randomization-inference p -values from 999 reassignments within strata/constituency at the assignment unit.

Table 65: Attrition, IPW, and Lee-Bound Checks

Experiment	Endpoint	T attr.	C attr.	Adj. diff.	OLS ITT	IPW ITT	Lee bounds
Country A	T3 endline	0.229	0.314	-0.075 (0.045)	0.031 (0.053)	0.028 (0.053)	[-0.023, 0.154]
Country B	endline	0.340	0.341	0.000 (0.062)	-0.212 (0.135)	-0.214 (0.135)	[-0.212, -0.212]
Country C	T2 ETE	0.309	0.323	0.001 (0.034)	0.146 (0.082)	0.137 (0.082)	[0.146, 0.180]
Country C	T3 ETE	0.350	0.490	-0.135 (0.084)	0.105 (0.132)	0.111 (0.130)	[-0.126, 0.298]

Notes: Attrition is missing outcome for the listed endpoint. Adjusted attrition differences use the same fixed effects as the corresponding outcome model. OLS ITT is the preferred endpoint regression. IPW estimates weight observed outcome regressions by the inverse predicted response probability from a logit on treatment, baseline score, and design fixed effects. Lee bounds trim the lower-attrition arm to equalize response rates and then re-estimate the covariate-adjusted endpoint regression. Standard errors in parentheses are clustered at the preferred assignment unit.

Table 66: Analysis Status and Multiplicity Disclosure

Family	Estimate	Status	Coef.	SE	Raw p	BH q
Primary endpoints	Country A primary ITT	Primary	0.031	(0.053)	0.564	0.564
Primary endpoints	Country B primary ITT	Primary	-0.212	(0.135)	0.120	0.359
Primary endpoints	Country C primary ITT	Primary	0.093	(0.133)	0.492	0.564
Track-position	Country A: lower-track effect	Heterogeneity	0.046	(0.137)	0.737	0.962
Track-position	Country A: higher-track differential	Heterogeneity	-0.023	(0.147)	0.877	0.962
Track-position	Country A: higher-track effect	Heterogeneity	0.023	(0.044)	0.597	0.962
Track-position	Country B: lower-track effect	Heterogeneity	-0.019	(0.181)	0.915	0.962
Track-position	Country B: higher-track differential	Heterogeneity	-0.328	(0.201)	0.106	0.239
Track-position	Country B: higher-track effect	Heterogeneity	-0.348	(0.154)	0.027	0.120
Track-position	Country C: lower-track effect	Heterogeneity	0.007	(0.152)	0.962	0.962
Track-position	Country C: higher-track differential	Heterogeneity	0.239	(0.098)	0.020	0.120
Track-position	Country C: higher-track effect	Heterogeneity	0.246	(0.129)	0.065	0.195
Country A payoff diagnostics	Treat x mover interaction	Mechanism	-0.000	(0.082)	0.996	0.996
Country A payoff diagnostics	Predicted assignment gain: Treat x gain (1 SD)	Mechanism	0.008	(0.068)	0.908	0.996
Country A payoff diagnostics	Top assignment-gain tercile ITT	Mechanism	-0.004	(0.056)	0.940	0.996
Country A payoff diagnostics	Treat x top assignment-gain tercile	Mechanism	-0.036	(0.089)	0.686	0.996

Notes: The table reports Benjamini-Hochberg false-discovery-rate q -values within each displayed family. The primary-endpoint family contains the country-specific ITTs that anchor the reduced form. The track-position family uses the primary endpoint in each country and the support-aware Country C two-group collapse. The Country A payoff-diagnostic family contains mechanism tests used to ask whether students with larger predicted assignment gains benefited more. Multiplicity adjustments are reported for transparency; the paper’s main conclusions rely on the pattern of estimates, implementation evidence, and structural benchmarking rather than on any single adjusted rejection.

Table 67: Country C ITT Robustness Across Specifications

	T2 MTE	T2 ETE	T3 MTE	T3 ETE
Baseline + FE + BL control	-0.042 (0.101)	0.146* (0.082)	0.071 (0.107)	0.105 (0.132)
No baseline control (FE only)	-0.114 (0.103)	0.090 (0.109)	0.051 (0.119)	0.017 (0.138)
Wild cluster bootstrap p value	0.695	0.132	0.555	0.423
Permutation p value	—	—	—	0.493
N	1898	2038	1463	1661

Notes: Primary endpoint is T3 ETE (column 4). Wild bootstrap uses 999 replications. Permutation p values use 999 reassignments of school treatment within constituency.

Table 68: Country C Lee Bounds by Assessment Wave

	T2 MTE	T2 ETE	T3 MTE	T3 ETE
Control attrition rate	0.366	0.323	0.484	0.490
Treatment attrition rate	0.378	0.309	0.530	0.350
Difference (T-C)	0.012	-0.014	0.046	-0.140
OLS ITT	-0.042	0.146	0.071	0.105
Lee lower bound	-0.074	0.146	-0.100	-0.126
Lee upper bound	-0.042	0.180	0.071	0.298

Notes: Attrition is missing outcome in each wave-specific column. Lee bounds trim the lower-attrition arm to match attrition rates.

Table 69: Treatment Effects With and Without Class-Size Controls

	Country A		Country B	
	(1) No CS	(2) With CS	(3) No CS	(4) With CS
Lower track (Treatment)	0.046 (0.137)	0.043 (0.138)	-0.019 (0.181)	-0.108 (0.180)
T $\times$ Upper	-0.023 (0.147)	-0.023 (0.148)	-0.328 (0.201)	-0.237 (0.189)
Upper track total	0.023 (0.044)	0.021 (0.044)	-0.348** (0.154)	-0.345** (0.151)
Class size		-0.001 (0.002)		0.003 (0.002)
N	4954	4954	3154	3154

Notes: Odd columns reproduce the track-interaction specification without class size controls. Even columns add realized reading class size as a control. Upper track total is the sum of the Treatment and T $\times$ Upper coefficients. All columns control for EB ability and strata FE. Standard errors are clustered at the school level in Country A and at the school grade group level in Country B. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 70: Country C: Treatment Effects With and Without Class-Size Controls (T3 ETE)

	(1) No CS	(2) With CS
Lower track (Treatment)	0.043 (0.155)	-0.055 (0.153)
T $\times$ Upper	0.194 (0.116)	0.405** (0.163)
Upper track total	0.236* (0.132)	0.350** (0.151)
Class size		0.011 (0.005)
N	1611	1611

Notes: Upper = higher group under the preferred support-aware two-group collapse. Upper track total = Treatment + T $\times$ Upper. Controls: baseline score, strata FE, grade FE. SEs clustered at school. ***, **, and * indicate 1%, 5%, and 10%.

Table 71: Robustness to Score Ceiling Effects (Country A)

	(1) OLS	(2) Tobit	(3) Trimmed OLS
Treatment	0.031 (0.053)	0.040 (0.058)	0.022 (0.057)
N	4954	4954	4471
Right-censored obs.		176	

Notes: Column (1) reproduces the baseline OLS ITT on standardized endline scores. Column (2) estimates a Tobit model on the raw endline composite (upper limit = 50 points), controlling for raw baseline score and strata FE; the coefficient and SE are divided by the control-group SD of raw endline scores for comparability with column (1). Column (3) re-estimates OLS after dropping students in the top baseline score decile. Columns (1) and (3) control for EB ability and strata FE; column (2) controls for raw baseline score and strata FE. Standard errors are clustered at the school level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 72: Country C Robustness to Ceiling Effects (T3 ETE Math)

	(1) OLS	(2) Tobit	(3) Trimmed OLS
Treatment	0.040 (0.150)	0.030 (0.155)	0.040 (0.150)
N	1661	1661	1661
Right-censored obs.		72	

Notes: Column (2) is Tobit on raw percent score with upper limit 100, normalized by control-group SD. Column (3) drops top base-line decile.

Table 73: Treatment Effect on Score Variance

	Country A	Country B
<i>Panel A: Aggregate variance</i>		
Control	0.988	0.921
Treatment	0.915	0.997
<i>Panel B: Treatment on total variance</i>		
Treatment	0.038 (0.073)	0.054 (0.210)
<i>Panel C: Treatment on within class variance</i>		
Treatment	-0.228*** (0.052)	0.137* (0.069)
<i>Panel D: Treatment on between class variance</i>		
Treatment	0.239*** (0.050)	-0.082 (0.199)
N	4954	3154

Notes: Panel A reports raw variance of standardized endline scores by treatment arm. Panels B–D regress individual-level squared deviations on treatment, controlling for EB ability and strata FE. Total variance uses deviations from the sample grand mean; within class variance uses deviations from realized reading classroom means; between class variance uses the squared deviation of the classroom mean from the grand mean. Standard errors are clustered at the school level in Country A and at the school grade group level in Country B. ***, **, and * indicate significance at 1%, 5%, and 10%.

Symbol	Definition
θ_i	Latent ability of student i
s_i	Observed baseline diagnostic score

$\mu_{g,m}$	Mean latent ability in grade g and setting m
$\rho_{m,g}$	Signal quality in grade g and setting m
a_i	Posterior expectation of ability given $(s_i, g(i))$
K_m	Number of tracks in setting m
$I_{k,m}$	Intended instructional target of track k in setting m
k_i^*	Intended track assignment under the cutoff rule
k_i	Realized track assignment after implementation frictions
ω_m	Assignment execution in setting m (summary of regime mixture and misclassification)
R_j	School-level implementation regime
$p_{m,r}$	Probability of implementation regime r in setting m
$M_{m,r}(k, \ell)$	Probability that intended track k is realized as track ℓ under regime r
τ_m	Assignment-channel delivery fidelity in setting m
$D_i(a)$	Delivered instructional content under regime a
$J_{c,m}$	Fallback classroom-responsive content in classroom c
$P_i(a)$	Leave-self-out mean peer ability under regime a
$r_i(a)$	Within-class percentile rank under regime a
$S_i(a)$	Composite social object, $P_i(a) + \omega_r r_i(a)$
$q_{c,m}$	Productive-time shifter for classroom c in setting m
N_c	Class size
V_c^g	Within-class grade dispersion
λ	Assignment-payoff scale in the estimable production map
φ	Strength of the composite social channel before scaling by $(1 - \tau_m)$
ω_r	Rank weight inside the composite social object
δ_m	Setting-specific baseline productivity shifter
χ_N	Effect of class size on productive time
χ_V	Effect of grade dispersion on productive time
Θ	Vector of structural parameters

Table 76: Local Identification of Stage-4 Production Parameters

Parameter	Role	Sensitivity norm	Largest local leverage
λ	Assignment-payoff scale	0.209	Country A: ITT
φ	Social-channel scale	2.132	Country A: Peer/rank
ω_r	Rank weight	0.717	Country A: ITT
χ_N	Class-size pressure	0.295	Country C: ITT
χ_V	Grade-dispersion pressure	0.165	Country C: Dispersion
α	Delivery activation	0.523	Country A: ITT
κ	Sorting compression	28.288	Country A: Dispersion

Notes: The sensitivity norm is the Euclidean norm of finite-difference derivatives of weighted fitted stage-4 moments with respect to the raw optimization parameter at the preferred estimate. “Largest local leverage” reports the target moment with the largest absolute weighted derivative. The table is a local diagnostic, not an additional identifying assumption.

Table 77: Stage-4 Production Parameter Uncertainty

Parameter	Role	Point	5th–95th pct.
λ	Assignment-payoff scale	0.400	[0.400, 0.404]
φ	Social-channel scale	0.192	[0.155, 0.216]
ω_r	Rank weight in social composite	-0.276	[-2.000, 2.000]
χ_N	Class-size pressure	0.033	[0.008, 0.183]
χ_V	Grade-dispersion pressure	0.022	[0.009, 0.054]
α	Delivery activation	4.615	[1.206, 10.025]
κ	Sorting compression	1.649	[1.614, 1.649]

Notes: Percentiles use the same 49 converged parametric re-estimation draws of the stage-4 target moments used for the conditional counterfactual intervals. Blocks 1–3 primitives are held fixed, so intervals describe uncertainty in the production block conditional on measured $(\hat{\rho}, \hat{\omega}, \hat{\tau})$.

Table 78: Social-Channel Parameter Sensitivity

Spec.	ω_r	Peer RMSE	Max ITT/SE	High-input	Country C $G + \tau$
Preferred	-0.276	0.518	0.282	+0.186	+0.199
No rank weight	0.000	0.520	0.275	+0.180	+0.191

Notes: The second row re-estimates the stage-4 production block after fixing the rank weight inside the social composite to zero. Peer/rank RMSE is the root mean squared weighted error across the three peer/rank diagnostic targets. Max ITT/SE is the largest absolute fitted-minus-target ITT residual divided by the corresponding study-specific standard error. The diagnostic asks whether the main assignment-delivery counterfactual relies on separately estimating a rank component.

Table 74: Country C Treatment Effect on Score Variance

	T2 MTE	T2 ETE	T3 MTE	T3 ETE
<i>Panel A: Aggregate variance</i>				
Control	0.761	0.707	0.888	0.684
Treatment	0.722	0.664	0.952	0.555
<i>Panel B: Treatment on total variance</i>				
Treatment	-0.035 (0.075)	-0.092 (0.103)	-0.123 (0.142)	-0.127 (0.086)
<i>Panel C: Treatment on within class variance</i>				
Treatment	-0.041 (0.045)	-0.008 (0.063)	0.052 (0.065)	-0.003 (0.043)
<i>Panel D: Treatment on between class variance</i>				
Treatment	0.024 (0.087)	-0.109 (0.070)	-0.166 (0.105)	-0.146* (0.081)
N	1898	2038	1463	1661

Notes: Treatment classes are school by `group_id`; control classes are school by grade. Controls: baseline score, strata FE, grade FE.

Table 79: Stage-4 Objective and Optimizer Diagnostics

Diagnostic	Value	Share (%)	Interpretation
<i>A. Objective decomposition</i>			
Moment fit	0.956	57.5	Weighted distance between targeted moments and model.
ITT moments	0.152	9.1	Contribution from the three country ATE targets.
Peer/rank moments	0.804	48.4	Contribution from the composite social-channel diagnostics.
Dispersion moments	0.000	0.0	Contribution from sorting-compression targets.
Assignment-payoff slope	0.000	0.0	Contribution from Country A's revealed assignment-payoff target.
Setting residual prior	0.546	32.8	Penalty for using country residuals to interpolate ITTs.
Auxiliary scale restrictions	0.160	9.6	Raw-parameter norm plus weak scale priors.
Hard comparative-static penalties	0.000	0.0	Zero means no acceptance inequality is binding.
<i>B. Numerical optimizer checks</i>			
Optimizer starts		25	Multi-start L-BFGS-B runs.
Converged starts		25	Starts reporting optimizer convergence.
Near-optimal starts		21	Starts within 10^{-3} of the best objective.
Objective range	1.662–1.684		Range across converged starts.
High-input ATE range	+0.180–+0.187		Range across converged starts.

Notes: Panel A decomposes the preferred stage-4 criterion at the optimum. Objective shares are relative to the total objective. The hard comparative-static penalties are the inequality penalties used to rule out counterfactual mappings that violate the model's monotonicity and ordering restrictions. Panel B reports multi-start optimizer diagnostics; the high-input ATE is the fully idealized assignment-value, execution, and delivery counterfactual.

Table 80: Stage-4 Normalizations

Term	Objective term	Preferred setting	Obj.	Role
Setting residual shrinkage	$\sum_m (\delta_m / 0.25)^2$	prior SD 0.25	0.546	Prevents country intercepts from mechanically interpolating ITTs.
Raw-parameter norm	0.05 mean($\tilde{\Theta}^2$)	weight 0.05	0.154	Numerical scale restriction on transformed parameters.
Assignment-payoff scale	$10[(\lambda - 0.4)/0.2]^2$	center 0.40; scale 0.20	0.000	Sets the preferred scale for high-fidelity assignment payoffs.
Social-channel scale	$2[(\varphi - 0.2)/0.15]^2$	center 0.20; scale 0.15	0.006	Keeps peer/rank scale in a plausible range.
Comparative-static penalties	monotonicity and ordering inequalities	hard penalties	0.000	Rules out counterfactual mappings that violate the theory.

Notes: The table reports the non-moment terms in the preferred stage-4 minimum-distance objective. “Obj.” is the contribution at the preferred estimate. The assignment-payoff and social-channel rows are scale normalizations; Appendix Table 20 reports sensitivity to relaxing them.

Table 81: Sensitivity to Setting-Residual Discipline

Specification	Residual prior SD	Max $ \delta_m $	Max ITT error / SE	High-input ATE
Strict	0.15	0.093	0.557	+0.181
Preferred	0.25	0.134	0.282	+0.186
Loose	0.50	0.186	0.097	+0.190
Unrestricted	none	0.500	0.003	+0.194

Notes: Each row re-estimates the stage-4 production block holding $(\hat{\rho}, \hat{G}, \hat{\omega}, \hat{\tau})$ fixed. “Max ITT error / SE” is the largest absolute fitted-minus-target ITT residual divided by the corresponding study-specific standard error. The high-input ATE sets assignment value G to Country A’s estimate, $\omega = 0.98$, and $\tau = 0.95$. The unrestricted specification removes residual shrinkage and hits the δ_m bound, so it is reported as a fit diagnostic rather than the preferred model.

Table 82: Influence of Stage-4 Target Blocks

Specification	Omitted target	Max ITT/SE	Country A high- τ	Country C $\rho + \tau$	High-input
Preferred	None	0.282	+0.152	+0.199	+0.186
Drop Country A ITT	Country A ITT	0.602	+0.121	+0.199	+0.157
Drop Country B ITT	Country B ITT	1.427	+0.145	+0.195	+0.177
Drop Country C ITT	Country C ITT	2.813	+0.157	−0.123	+0.189
Drop peer/rank	Peer/rank moments	0.287	+0.125	+0.161	+0.157
Drop assign-payoff	Country A assignment-payoff slope	0.283	+0.156	+0.205	+0.187
Drop dispersion	Within-class dispersion moments	0.282	+0.152	+0.199	+0.186

Notes: Each non-preferred row re-estimates the stage-4 production block after omitting the indicated target block, holding the stage-1 to stage-3 primitive estimates fixed. “Max ITT/SE” evaluates fitted observed-country ATEs against the harmonized study-specific ITT estimates, even when an ITT target is omitted from estimation. The exercise is a local influence diagnostic: the preferred specification remains the full target set.

Table 83: Setting-Level Influence of Stage-4 Targets

Spec.	Omit	Holdout	Max fit	K high- τ	NG $G + \tau$	High-input
Preferred	None		0.282	+0.152	+0.199	+0.186
Drop Country A	Country A	1.089	1.089	+0.111	+0.218	+0.152
Drop Country B	Country B	1.582	1.582	+0.129	+0.175	+0.160
Drop Country C	Country C	2.828	2.828	+0.152	-0.132	+0.183

Notes: Each non-preferred row re-estimates the stage-4 production block after omitting all targeted stage-4 moments from the indicated setting, holding the stage-1 to stage-3 primitives fixed. Held-out ITT/SE is the absolute difference between the model-predicted observed-cell ATE and the harmonized reduced-form ITT for the omitted setting, divided by that ITT’s standard error. The exercise is a setting-level influence diagnostic, not an alternative preferred specification.

Table 84: Sensitivity to High-Input Primitive Benchmarks

Scenario	G	ω	τ	High-input ATE
Conservative joint	0.320	0.95	0.90	+0.096
Lower G only	0.320	0.98	0.95	+0.120
Lower ω only	0.533	0.95	0.95	+0.181
Lower τ only	0.533	0.98	0.90	+0.150
Preferred	0.533	0.98	0.95	+0.186

Notes: Each row evaluates the high-input counterfactual in the Country A production environment while holding the estimated production parameters fixed. G is the signed assignment-value primitive used in the payoff term. The conservative joint row uses the lower positive observed assignment-value benchmark, $\omega = 0.95$, and $\tau = 0.90$. The preferred row uses the paper’s main high-input benchmark, $G = \max_m \hat{G}_m$, $\omega = 0.98$, and $\tau = 0.95$.

Table 85: Sensitivity to Primitive-Estimation Uncertainty

Scenario	Point	5th pct.	Median	95th pct.
Country A high delivery	+0.152	+0.152	+0.152	+0.152
Country C high delivery only	+0.086	+0.083	+0.086	+0.089
Country C high assignment value + delivery	+0.199	+0.182	+0.199	+0.217
Country C all three	+0.214	+0.214	+0.214	+0.214
Fully high-input cell	+0.186	+0.186	+0.186	+0.186

Notes: This table propagates uncertainty in the constructed assignment-value primitive, delivery fidelity, and assignment execution while holding the preferred stage-4 production mapping fixed. Assignment-value draws use the student-level predicted mismatch reductions; delivery-fidelity draws use the process-moment variances; Country C assignment-execution draws use a 0.08 SD implementation-error scale, while deterministic Country A and Country B execution is held fixed. The table is a primitive-uncertainty sensitivity check, not a full-system bootstrap.

Table 86: Combined Primitive and Production Uncertainty

Scenario	Point	5th pct.	Median	95th pct.
Country A high delivery	+0.152	+0.027	+0.126	+0.201
Country C high delivery only	+0.086	−0.108	+0.081	+0.295
Country C high assignment value + delivery	+0.199	−0.035	+0.182	+0.394
Country C all three	+0.214	−0.027	+0.188	+0.407
Fully high-input cell	+0.186	+0.057	+0.166	+0.221

Notes: This diagnostic combines uncertainty in the assignment-value, delivery-fidelity, and execution primitives with the stage-4 production-parameter re-estimation draws. Primitive draws use the same sampling rules as Table 85; production-parameter draws are sampled from the converged stage-4 target-moment re-estimations used in Table 77. The exercise is an approximate combined-uncertainty check, not a nonparametric bootstrap of the three original experiments.

Table 87: Sensitivity to Delivery-Activation Functional Form

Spec.	α	Obj.	Assign-payoff	Max ITT/SE	K high- τ	High-input
Linear	1.00	3.876	+0.100	0.318	+0.120	+0.127
Quadratic	2.00	2.154	+0.053	0.297	+0.152	+0.168
Cubic	3.00	1.743	+0.027	0.287	+0.160	+0.184
Estimated	4.61	1.662	+0.009	0.282	+0.152	+0.186
Steeper	6.00	1.661	+0.003	0.281	+0.139	+0.179

Notes: Each row re-estimates the stage-4 production block under the indicated delivery-activation exponent in τ^α , holding the stage-1 to stage-3 primitives fixed. The preferred row estimates α ; the other rows fix it. “Obj.” is the full stage-4 minimum-distance criterion, including moment fit and regularization terms. “Assign-payoff fit” is the model-implied Country A coefficient on treatment interacted with the predicted assignment-gain index, whose data target is +0.008. The table shows the fit–extrapolation tradeoff behind the delivery-activation restriction.

Discussion and Conclusion

The three experiments do not give a single verdict on ability grouping. None of the observed implementations produced a positive average effect. The sources of the weak effects differ across countries. Country B replaced grade with an assessment with low predictive content. Country A formed the groups and reduced predicted mismatch, but sorting at observed delivery did not raise achievement. Country C was designed as the high-input case, but group execution was incomplete and the curriculum carrying the assignment contrast was delivered in only 26% of scheduled lessons.

The calibrated benchmark maps these observed cases into the missing high-input case. Under Country A-level assignment value, near-complete assignment execution, and high assignment-channel delivery fidelity, the preferred predicted ATE is about +0.19 SD. The benchmark gives scale; the sensitivity exercises give the more stable implication. Raising assignment execution has limited predicted payoff when delivery is low or when the placement rule does not improve match. Raising predictive content has limited predicted payoff when the assigned curriculum is not delivered. The return to cross-grade grouping appears when students are assigned to content consistent with their preparedness and that content is taught.

This interpretation separates two margins in the tracking literature. The equity tradition, associated with Oakes (1985) and Gamoran (1992), documents stigma, lower teacher expectations, weaker instructional quality in lower tracks, and widening inequality. The efficiency tradition, including the Joplin Plan evidence and the Country A tracking experiment in Duflo et al. (2011), documents gains when grouping changes the instruction students receive. These literatures need not disagree about the realized effects of any particular grouping policy. They emphasize different parts of the same production process: the signal used to form groups, whether assignments are revised or fixed, what teachers do with the groups, and whether lower tracks receive content that builds the skills students have not yet mastered.

In this paper, negative effects arise when the placement signal has low predictive content or when the assigned curriculum is not delivered. Those mechanisms are consistent with the equity literature, but they are not mechanical consequences of matching students to instructional level. A lower group can receive weaker instruction and lower expectations; it can also receive foundational content that grade-level instruction was skipping. Which case obtains depends on measurement,

reassessment, instructional quality, and organizational design.

Scripted instruction changes these margins. It can make misassignment more costly because the teacher has less discretion to compensate for placement errors. It can also limit some sources of across-track variation if the lower-track script is aligned to foundational skills and delivered at comparable fidelity and quality. The experiments do not measure stigma, teacher expectations, or student identity directly, so they cannot adjudicate those channels. They show instead that equity and efficiency depend on the same implementation objects: the predictive content of the placement signal, the maintenance of groups, and the delivery and quality of content across tracks.

The welfare comparison also includes the grade-based counterfactual. Much of the anti-tracking literature compares the stigma of a lower group to an ideal heterogeneous classroom. In the settings studied here, a grade-based classroom can also expose students to persistent mismatch. A child who cannot read the text or follow the mathematics lesson is assigned by grade to content she cannot learn from. The comparison therefore includes both the psychological cost of a track label and the psychological and human-capital cost of repeated failure in grade-level instruction. Dynamic skill complementarities make the latter cost potentially persistent: missing foundations today lowers the productivity of tomorrow's instruction (Cunha and Heckman, 2007).

The policy decision can be stated as four quantities. How informative is grade? How much does the placement assessment improve predicted instructional match relative to grade? Can the school form and maintain the intended groups? Does each group receive content at the assigned level and of comparable quality? The predicted return to cross-grade grouping is limited when any one of these quantities is low. The binding margin may be measurement, management, or instruction, and the return to improving one margin depends on the others.

The same logic applies outside education. Assignment policies often begin with measurement, but measurement changes outcomes only through the production environment that follows the assignment. A more informative signal raises output when the organization can execute the rule and when the assigned environment changes the input people receive. In these experiments, assessment, group formation, and instructional delivery enter together. The organizational problem is not only to know more about students, but to convert that information into instruction.

Cross-grade grouping returns to an old instructional idea: teach students at the level they are prepared to learn. The experiments show why that idea does not reduce to sorting. Grade-based

assignment can be mismatched to skill, so an assessment-based assignment rule can have value. But the rule must predict the skill that determines instructional match, the school must act on it, and instruction must make the assignment consequential. The next empirical question is whether school systems can achieve those conditions at scale.

Transparency and replication. The three experiments were not designed as one pooled experiment. The paper therefore separates country-specific primary ITTs from harmonized pooled estimates, support-aware Country C grouped estimates, mechanism diagnostics, and the calibrated structural benchmark. Appendix Table 66 reports the status of displayed estimate families and false-discovery-rate q-values. The replication package contains the Stata and Python code that generates the tables, figures, structural outputs, and compiled manuscript from the repository-local pipeline.

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